

TECHNICAL MANUSCRIPT

Obstructions in Shared-Probe HSS Approximation

Counterexamples, failed estimates, repaired gaps,
and the remaining barriers

Research snapshot: 20 September 2026

Status. The general conjecture remains open in this research record. This manuscript records what failed, why it failed, and the exact scope of each obstruction.

Prepared from the supplied proof manuscript and the subsequent proof, audit, and numerical records.

Real matrices. Fixed inputs. Exact arithmetic in theorem statements.
Squared Frobenius error and explicit conditioning conventions.

Abstract

This manuscript consolidates the obstructions encountered in an attempt to obtain an $O(k)$ matrix–vector query algorithm for quasi-optimal HSS approximation. The desired guarantee is an expected squared Frobenius error bounded by $CL\text{OPT}_k(A)$, with final HSS rank at most k and polynomial additional computation. Neither a proof nor a disproof of that conjecture has been obtained.

The record distinguishes failures of proof estimates from failures of specified algorithms. It includes adaptive Gaussian conditioning, inverse-moment growth on exact inputs, noncommutative and matrix-embedding obstructions in refitting, actual noisy-input counterexamples for earlier basis learners, failure of screened residual estimates after dropping the final projection, and failures of global interpolation selectors even with exact leaf information. It also records why local independent-source theorems, independent pilots, and finite numerical tests do not yet imply the general shared-probe result. Historical refitting obstructions are presented with their resolutions: the independent fixed-space refit now has a depth-uniform factor 1607.

Later counterexamples distinguish the old full-range learner, separately trained two-view hierarchies, and the current synchronized learner. Negative results for the first two do not refute the third. The concluding status map identifies the remaining adaptive signed-error estimate and the separate efficient rank-reduction requirement. New observations that had not completed independent review when research stopped are explicitly segregated from established results.

Scope and conventions

This is a technical synthesis of `hss_complete_proof_chain.tex` and the subsequent local proof and audit notes, frozen on 20 September 2026 at the user’s instruction to stop research. It is not a claim that every retained argument has received external peer review. Here “reviewed” or “audited” means that a corresponding internal mathematical audit was recorded. The supplied original manuscript has not been overwritten.

The motivating public benchmark is Amsel et al. [1]: expected squared Frobenius error $O(L) \text{OPT}_k(A)$ using $O(kL)$ queries. The open improvement seeks to remove the depth factor from the query count, while retaining the approximation guarantee, output rank, and computational requirement. Levitt–Martinsson’s reused-sketch construction is relevant background, but its discussion of coarse-level sketch quality does not provide the missing all-input approximation theorem [2]. This manuscript makes no exhaustive literature-absence claim.

All probabilistic assertions concern a fixed input chosen before the random probes unless explicitly stated otherwise. A data-dependent competing interpolant is permitted in an argument about the failure of an estimator; it is not silently reinterpreted as a fixed adversarial input. Positive-probability event constructions are distinguished from lower bounds whose probabilities remain uniform under a parameter limit.

We use $\text{OPT}_k(A) = \inf_{B \in \mathcal{H}_k} \|A - B\|_F^2$ and write L for tree depth, counting the nonroot cut levels under the local convention. Physical leaf sizes differ between candidate algorithms and are specified with each result. Node projectors onto coordinate sets and projectors onto retained learned spaces are different operators; they are disambiguated locally.

The companion manuscript, *The Current Proof Route for Shared-Probe HSS Approximation*, specifies the surviving synchronized construction and its exact outstanding estimate. The present document groups repeated attempts by mathematical obstruction rather than reproducing a chronological conversation.

Status vocabulary. A *counterexample* refutes the particular quantified claim stated beside it. A *gap* is a premise still needed by an otherwise valid implication. A *repaired obstruction* explains why an earlier argument failed and what later theorem replaces it. A *numerical diagnostic* is finite evidence under its reported sampling design. A *working observation* was not promoted to an audited theorem before the stop point.

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Original proof-chain obstructions

1.1 Obstructions in the original manuscript: a logical taxonomy

This chapter consolidates the negative results and proof limitations in `hss_complete_proof_chain.tex`. Its purpose is to specify what each obstruction actually excludes. A counterexample to a proposed potential, a failure of a particular learned space, and an information theoretic lower bound are different statements. The original manuscript contains examples of the first two kinds, but no impossibility theorem for the general $O(k)$ -query HSS approximation problem.

Throughout, A is fixed before the probes, $\mathcal{H}_k$ is the HSS rank- k class on the prescribed finite coordinate tree, and

$$e_k(A) = \inf_{B \in \mathcal{H}_k} \|A - B\|_F^2.$$

For a learned frozen coefficient space $\mathcal{S}$, write $d_{\mathcal{S}}(A)^2 = \|(I - P_{\mathcal{S}})A\|_F^2$. When $\mathcal{S} \subseteq \mathcal{H}_r$, a theorem relative to $d_{\mathcal{S}}(A)$ is a statement about that particular learned space, not automatically a theorem relative to $e_k(A)$. In particular, $d_{\mathcal{S}}(A)^2 \geq e_k(A)$ when $\mathcal{S} \subseteq \mathcal{H}_k$; the inequality points in the wrong direction for replacing the fixed-space comparator in an upper bound.

Status convention. We distinguish *counterexamples*, which disprove a stated universal claim; *missing premises*, which identify a necessary input to an otherwise valid implication; and *repaired historical gaps*, whose resolution appears later in the same original manuscript. The appendix labelled `base:new:route-history` records 66 routes, and `base:new:obstruction-ledger` records their logical scopes. Some of its final paragraphs preserve an earlier, unresolved status for fixed-space refitting. That historical status is superseded by Chapters *Frozen-stream feedback*, *Coherent drift*, and *Uniform and near-optimal fixed-space refitting* (`ch:feedback`, `ch:drift`, `ch:uniform`). The established endpoint is the factor 1607 with $8k$ independent refit queries. The basis-learning premise remains separate.

1.1.1 Comparators, query accounting, and exactness are separate interfaces

Fresh queries at every level supply a legitimate comparison algorithm, but an $O(kL)$ -query proof does not become an $O(k)$ -query reuse proof by omitting the freshness hypothesis. Likewise an $O(L)$ learning factor and an $O(L)$ coefficient-refitting factor compose to $O(L^2)$, not to the desired $O(L)$. These accounting restrictions are explicit in the original query ledger, `base:new:ledger`.

Exact recovery on fixed HSS inputs is also weaker than residual-relative stability. An exact trajectory may be discontinuous across a rank-changing stratum, and its inverse constants may be unbounded near that stratum. Canonical completion removes arbitrary choices of zero singular vectors; it does not provide a quantitative perturbation theorem. The original raw learner's exact-input result and its first two compression estimates therefore do not supply an all-depth approximation induction (`base:sec:exact-recovery`, `base:sec:finite`, `base:sec:second`).

For a known bicriteria output C , exact offline HSS rank reduction is a valid query-only operation: if $\hat{B}$ minimizes $\|C - B\|_F$ over $\mathcal{H}_k$, comparison with a best B_* for A gives

$$\|A - \hat{B}\|_F \leq 2\|A - C\|_F + \sqrt{e_k(A)}.$$

This does not assert polynomial-time computability or authorize replacing the exact minimizer by a practical recompression routine with a different factor. Nor does a lower bound for the unprocessed learned-space candidate automatically persist after arbitrary nonlinear postprocessing. These qualifications are part of `ch:bicriteria` and `ch:synthesis`.

1.2 Gaussian beginnings do not give Gaussian adaptive regressions

1.2.1 What ambient Gaussian compression really does prove

Let $G \in \mathbb{R}^{d \times s}$ be standard Gaussian, $s > d + 1$, and let $W \in \mathbb{R}^{d \times r}$ be any measurable isometry, even one chosen from G . The valid pathwise comparison is

$$(W^T G G^T W)^{-1} \preceq W^T (G G^T)^{-1} W, \quad \mathbb{E} \|(W^T G)^\dagger\|_F^2 \leq \frac{d}{s - d - 1}.$$

The inverse is controlled by the *ambient* dimension d , not merely the retained dimension r . The proof is a Schur-complement comparison followed by the Gaussian inverse trace formula. Thus adaptivity alone does not invalidate every inverse estimate; it invalidates silently replacing an adaptively compressed design by a fresh r -row Gaussian matrix. The source is Lemma `base:lem:inverse-compression`.

1.2.2 Nested full rank can coexist with an infinite inverse moment

Proposition `base:prop:nested-obstruction` constructs a measurable binary hierarchy of rank- k orthogonal compressions of a square standard Gaussian $G \in \mathbb{R}^{8k \times 8k}$. Its cumulative isometry $W(G) \in \mathbb{R}^{8k \times 2k}$ has full-row-rank compressed design $W^T G$ almost surely, but

$$\mathbb{E} \|(W^T G)^\dagger\|_F^2 = \infty.$$

Choose a left singular vector u for the smallest singular value of G . At each binary stage retain the restrictions of u in the appropriate children, padding orthonormally as needed. The resulting nested range contains u , so compression retains the smallest eigenvalue of $G G^T$. The reciprocal squared distance of the last Gaussian row from the span of its predecessors has the law Z^{-2} with $Z \sim N(0, 1)$ and infinite mean. It lower-bounds the relevant inverse norm.

The input to this construction is the probe itself. It is not claimed to be the SVD hierarchy reached by the actual learner on a fixed A . Its conclusion is exact: initial Gaussianity, orthonormality and nesting alone are insufficient hypotheses. With more physical rows than probe columns, unrestricted adaptive nesting can even preserve a vector in the exact left kernel. The fractional-moment version is stronger. In Proposition `base:aug:prop:all-fractional`, let $G \in \mathbb{R}^{d \times s}$ with $d > s$, choose unit vectors $u_0 \in \ker G^T$ and u_1 in a largest left singular direction, and put

$$R = \|G\|_F, \quad \varepsilon = \exp(-R^4), \quad w = \sqrt{1 - \varepsilon^2} u_0 + \varepsilon u_1.$$

Then $w^T G$ is nonzero almost surely, while for every $p > 0$

$$\|(w^T G)^\dagger\|^{2p} \geq \exp(2pR^4) R^{-2p}, \quad \mathbb{E} \|(w^T G)^\dagger\|^{2p} = \infty.$$

The chi-density of R cannot offset the quartic exponential. This excludes a universal fractional inverse theorem for arbitrary adaptive compressions; it does not disprove a weak-tail theorem for the actual HSS rule.

1.2.3 Separate energy and inverse moments do not price their product

Example `base:ex:correlated-inverse` uses a standard Gaussian row g and $p_\varepsilon = \Pr\{\|g\| \leq \varepsilon\}$. Set

$$f(g) = p_\varepsilon^{-1/2} \mathbf{1}_{\{\|g\| \leq \varepsilon\}} \frac{g}{\|g\|}.$$

Although $\mathbb{E}\|f(g)\|^2 = 1$, the regression cost satisfies

$$\mathbb{E}|f(g)g^\dagger|^2 = \mathbb{E}[\|g\|^{-2} \mid \|g\| \leq \varepsilon] \geq \varepsilon^{-2}.$$

Even a finite marginal inverse moment does not justify factorizing this dependent product. The same issue survives replacing a conditional Wishart covariance by its expectation: inverse moments and nonlinear regression risks do not commute with that averaging. The correct Wishart interface in `base:new:wishart` first identifies the original conditional Gaussian law; it is not a freshness argument.

1.2.4 Stein corrections, orthogonal mixing, and ridge do not give a shortcut

The regularized integration-by-parts calculation in `base:sec:closure` separates a fresh regression cost from an adaptation derivative. Its sign is not universally favorable. If a projector selects the r weakest left singular directions of an m -row design, it selects the r largest inverse eigenvalues. Their sum is strictly larger than r/m times the full inverse trace, so the risk exceeds the fresh coefficient. This is a counterexample to a sign inference, not a description of the actual HSS selector. The zero-ridge limit additionally requires justified integrability; a regularized identity alone does not provide it.

Common right orthogonal mixing of a data/design pair, $(Y, \Omega) \mapsto (YO, \Omega O)$, leaves the regression and innovation Gram unchanged. It merely changes query coordinates and does not refresh the estimator. Positive ridge in the discrepancy has a different defect: it generally biases even exact inputs such as the identity, whose nullified innovations vanish. Exact recovery is a necessary requirement for a residual-relative theorem when $e_k(A) = 0$. These are the two modifications discussed in `base:sec:additional-obstructions`; neither settles the reuse problem.

1.3 Validation and global fitting: exact implications with missing premises

The validation route in `base:sec:validation` is a conditional amplification theorem. It can convert a suitable weak tail for the *actual* candidate error into an expected-error guarantee by making independent candidates and testing them with independent probes. It does not establish that weak tail. The distinction is visible even in one dimension: if $F = Z^{-2}$ for a standard normal Z , then $\Pr(F > t) \leq \sqrt{2/\pi} t^{-1/2}$; the minimum of three independent copies has finite expectation, whereas the minimum of two still does not. Conversely, a statement saying only that a candidate succeeds with probability 0.9 is insufficient. A loss equal to M with probability 0.9 and TM otherwise has

$$\mathbb{E} \min_{1 \leq i \leq a} F_i = M[1 + (T-1)10^{-a}],$$

which is unbounded in T for every fixed number a of candidates. Clipping or a fixed-probability success statement cannot substitute for tail control relative to the unknown optimum.

Training consistency has a separate uniformity defect. If both training widths are below the ambient dimension, choose $u \in \ker \Psi^T$ and $v \in \ker \Omega^T$. The rank-one matrix uv^T is invisible to

both sketches. This rules out a residual certificate uniform over all sketch-dependent candidates, even within HSS rank one. Because this invisible matrix is chosen after the probes, the example is not a fixed-input information lower bound.

1.3.1 Identifiability does not provide uniform nonlinear conditioning

The positive identifiability result in `base:con:identifiability` must be retained alongside its limitations. For a fixed HSS matrix whose prescribed cut ranks are saturated at k , sufficiently wide two-sided Gaussian observations (width at least $3k$ in the stated theorem) identify its cut spaces and cores almost surely. The competing HSS matrices may depend on the observations. A local smooth inverse exists on the corresponding full-rank, spectral-gap charts. Its radius and Lipschitz constant are random, input-dependent quantities; the argument does not integrate them to obtain uniform expected approximation.

Nonsaturated inputs behave differently. If a unit vector u lies in the common nullspace of Ω^T and Ψ^T , then I and $I - uu^T$ have identical observations. Their off-diagonal cut ranks are at most one, but $I - uu^T$ has smaller Frobenius norm. Thus a minimum-norm completion can fail on the exact identity input. The manuscript's lexicographic cut-rank selector addresses exact rank selection, not noisy stability.

The finite-family route in `ch:finite` has a different obstruction. Even leaf-block-diagonal members contain a Euclidean family of dimension proportional to Nk . A generic net therefore has logarithmic cardinality of that order, and generic finite-model selection brings an ambient-size penalty. This is a limitation of the net argument: those same diagonal blocks can be recovered with $2k$ structured forward products. Likewise, direct global least squares can be injective without a proved depth-uniform lower bound on the measurement map over the entire coefficient space.

1.3.2 A concrete failure of depth-weighted nuclear selection

Theorem `nuclear:thm:failure` is an actual selector counterexample, not merely a conditioning warning. Divide the root into two halves I, J of size n , let the two probe widths be q , and fix $B[I, J] = uv^T$ with constant unit vectors u, v ; all other blocks vanish. Let P project onto range Ψ_I and Q onto range Ω_J . The sketch-dependent competitor

$$C = Pu v^T + u v^T Q - Pu v^T Q = (Pu)v^T + (u - Pu)(Qv)^T$$

has rank at most two and the same two observations as B . Setting $a = \|Pu\|$ and $b = \|Qv\|$ gives

$$\|C\|_F^2 = a^2 + b^2 - a^2 b^2, \quad \mathbb{E}(a^2 + b^2) = 2q/n.$$

Hence $\|C\|_F^2 < 1/2$ on an event of probability at least $1 - 4q/n$.

For any nonnegative depth weights, not all zero, the sum of the nuclear norms of both orientations of all cuts at a given depth is $2\sqrt{m}$ for B , where m is the number of nodes at that depth. For C it is at most $2\sqrt{2m}\|C\|_F$. Thus the weighted tree nuclear penalty is strictly smaller for C on that event. For $k \geq 2$ and $n > 4q$, either exact completion by this penalty or least-squares fitting plus a positive multiple of it fails to return the fixed exact input B . The conclusion excludes this specified regularizer, with its stated depth weights; it does not exclude all global fitting objectives.

1.4 Signed quotient energy and the failure of scalar bookkeeping

In the original refit analysis, the innovation and mismatch state removes a common coefficient that has no effect on the next basis selection. Its energy may be written

$$V = \|E_Y\|_F^2 + \|E_Z\|_F^2 + \|\Delta\|_F^2.$$

Compression dissipates this energy exactly:

$$V' = V - \|U_\perp^T E_Y\|_F^2 - \|V_\perp^T E_Z\|_F^2 - \|\Delta_{22}\|_F^2.$$

At a merge, however, the only surviving interaction is signed:

$$V_{\text{parent}} = V_1 + V_2 - 2\langle F_Y, F_Z^T \rangle.$$

The scalar-weight experiment in `base:new:energy` is conclusive for its proposed class: compression requires a weight at least one, whereas the state $F_Z^T = -F_Y$ makes the merge require a weight at most one half. No single scalar weight contracts both maps on every admissible state.

Even a successfully contracting quotient is not an output theorem. The pair $Y = C\Omega$, $Z = C^T\Psi$ has zero quotient but can carry a nonzero common coefficient C . Emitted coefficients and the terminal root must be recorded separately. Similarly, inactive error is a reconstruction charge, not future active forcing; charging it at every later level repeatedly counts the same physical loss (`base:sec:global`, `base:new:ledger-interface`).

The reachable rank-two witness in `base:new:scalar-storage` shows that adding a scalar residual storage does not cure the problem. With $P = e_1 e_1^T$, $Q = I - P$, use the completion

$$\begin{pmatrix} P & -Q \\ Q & P \end{pmatrix}.$$

At $k = 2, q = 8$, the source state can have $Y = F = C = D = 0$ and $R = J_0 = e_1 e_1^T$. A fresh row ξ then produces $C' = \xi_2 e_1 e_2^T$, $F' = -\xi_2 e_2 g_1$, and $R' = (1 - \xi_2^2) e_1 e_1^T$, with $\mathbb{E}\xi_2^2 = 1/4$. The proposed scalar output-plus-storage inequality fails for every scalar weight, although the actual output Y' is zero. Directional matrix storage resolves this particular witness; it does not turn its restricted geometric hypotheses into a theorem for arbitrary learned bases.

1.4.1 Independent one-pass estimates are not feedback estimates

A centered one-pass map K_G can satisfy $\mathbb{E}\|K_G b\|^2 \leq \rho\|b\|^2$ for deterministic b , or for b independent of G . In the coupled equations $a = a_0 - K_G b$, $b = b_0 - K_H a$, the solved inputs are functions of both streams. Iterating the independent-input estimate is therefore an unjustified substitution.

Chapter `ch:halfsolve` supplies a finite-dimensional counterexample even with independent half solvers. Let S be the nilpotent shift on $\mathbb{R}^h$ and $P = \text{diag}(1, \dots, 1, 0)$. Independently choose ξ, η equal to ± 1 with respective total probabilities a, b , and zero otherwise; set $C = \xi S$, $D = \eta P$. Both maps are centered contractions with the appropriate one-pass second-moment bounds, and $CD = \xi\eta S$ is nilpotent. Nevertheless

$$\mathbb{E}\|(I - CD)^{-1} e_1\|^2 = 1 + ab(h - 1).$$

The same rare signs activate every power of the feedback loop; their probabilities do not multiply afresh at each traversal. A residual injection gives coupled physical error $a[1 + ab(h - 1)]$ while preserving the stated individual half-solve guarantees. This abstract obstruction does not prove failure of the specific HSS refit or of a positive global least-squares decoder.

1.5 Matrix Carleson control does not remove anticipative alignment

The restart theorem in `base:new:carleson` provides an adapted conditional future-response estimate. Its valid proof retains the full cross covariance tensor and deterministic mean transport (`base:con:restart-cone`). That is a repaired historical gap: vanishing of a scalar cross trace alone was insufficient to preserve the required covariance cone. The source decomposition also had to include discarded directions *inside* a nonleaf sibling. Example `base:con:interior-example` invalidates their omission. The repaired coarse off-chain budget remains valid, with the distinct $q - 2k$ and $q - 3k$ denominators kept in their respective modules.

These repairs do not imply a dimension-free embedding for an anticipatively selected terminal source. The distinction between an adapted restart and a source using future probe information is essential. The original embedding route yields logarithmic rank/time factors; an isotropy assertion alone cannot delete them.

1.5.1 The Haar-harmonic obstruction

The construction in `base:new:abstract-obstruction` takes $m \geq 4$, ambient dimension at least $2m$, and successive Haar orthonormal columns v_i . Define

$$f_n = \sum_{i \leq n} v_i, \quad u_j = \sum_{i \leq j} \frac{v_i}{j - i + 1}, \quad A_1 = 0, \quad A_n = u_{n-1} u_{n-1}^T.$$

The increments of f_n are conditionally centered and spherical. The weights are predictable and positive semidefinite; their norms are bounded by $\pi^2/6$. The manuscript checks the conditional future-tail bound using the Hilbert-matrix norm bound π . Nevertheless the embedding energy is

$$\sum_n \langle f_n, A_n f_n \rangle = \sum_{j=1}^{m-1} H_j^2, \quad H_j = \sum_{i=1}^j \frac{1}{i}.$$

The available variance budget is at most $4m$, while the normalized gain is at least $\log^2(m/2)/16$.

This disproves a dimension-free consequence of the listed coarse properties: conditional isotropy, positive predictable weights, bounded expected tails, and the corresponding budget are not enough. It is not an HSS counterexample. The orthogonal family grows with m , whereas the HSS path has a fixed query width. The example also shows why an abstract p^2 response-moment estimate cannot simply be replaced by an $O(p)$ estimate. Adding a decreasing quotient form does not fix the abstract implication, since the constant form I already satisfies that condition.

1.5.2 Frozen-forward gain can be arbitrarily large

The conditional amplifier in `base:new:amplifier` is an actual fixed-basis refit construction, but its conclusion is conditional. At $k = 1, q = 4$, choose M blocks and an odd $n \geq 4\pi^2 M$, put $c = 2 \cos(\pi/n) - 1$ and $s = \sqrt{1 - c^2}$, and alternate a run of n small column rotations with a release step. The relevant forward transfer is

$$\begin{pmatrix} c & 0 & s \\ -s & 0 & c \\ 0 & 1 & 0 \end{pmatrix}.$$

Its eigenvalues are $-1, e^{i\pi/n}, e^{-i\pi/n}$, so its n th power is $-I$ while c^n remains close to one. After a fixed warm-up, each release extracts a uniformly positive coefficient. The frozen-forward

response satisfies $W(G) \geq M/16$ at height $2 + M(n + 1)$, and continuity gives a positive-probability neighborhood with response at least $M/32$.

The neighborhood probability is not uniform in M . In fact the fully averaged independent unit-source gain in the same construction is bounded by $5/4$. Choosing still slower rotations suppresses the actual source excitation: the manuscript obtains a source-weighted joint average at most $14.402\beta/M$ while the conditional unit gain is at least $M/16$. Thus a uniform bound after freezing the whole forward stream is false; the unconditional physical theorem is not disproved. Near-protected covariance must be charged jointly with the response rather than separated from it.

1.5.3 Temporal parity and finite moment closure have limited scope

Immediate fresh-frame parity kills a current odd cross term, not a future weighted one. For the temporal kernel $K_{rs} = \mathbb{E}[T_r W T_s^T]$, the surviving delayed contribution for $r < s$ involves future readouts after time s . Exact rational calculations in `base:new:temporal` and `base:new:fourth`, using the original Haar law, exhibit nonzero delayed entries. For example, with the stated scalar 3–4–5 rotations,

$$K_{35} = \frac{2925051264}{95367431640625} > 0, \quad K_{45} = \frac{4766750208}{476837158203125} > 0.$$

They are finite-family certificates, not an all-depth stability theorem. Fixed polynomial degree and a finite covariance state do not give a spectral contraction; Haar integration also has signed coefficients, so the monomial transfer is not a positive scalar recursion.

Absolute-value path bounds lose another cancellation. A sequence of h overlaps each of size $1/\sqrt{h+1}$ has bounded sum of squares but an absolute sum of order $\sqrt{h}$. Bessel control therefore does not make the absolute ancestor sum bounded. Orthogonality of distinct initial sources removes cross-source terms but leaves same-source temporal returns.

1.6 Noncommutative refit estimates: the information that cannot be dropped

1.6.1 Source moments, covariance coercivity, and coherent rows

The Gaussian/gamma moment-generating-function envelopes in `base:new:source-moments` are inequalities for the original source. They do not identify its distribution or give stochastic or convex order. Likewise, auxiliary random signs used to take a square root of a response are an analysis device, not replacement probes. A tail bound for conditional expected risk is not automatically a tail bound for realized reconstruction error.

For the exact source covariance and geometric loss, `base:new:coercive` proves the sharp comparison

$$(1 - \rho)^2 \mathsf{L} \preceq \mathsf{C} \preceq \mathsf{L}, \quad \rho = \frac{k}{q - 2k}.$$

At $q = 4k$ the lower coefficient is $1/4$, attained by actual frames. Thus an arbitrarily large improvement cannot come solely from claiming that the exact covariance is much smaller than geometric loss. The comparison does not control how a dependent future response aligns with that covariance.

The fourth-moment method in `base:con:gram` keeps $\text{tr}[(N^T N)^2]$ rather than replacing it by a crude Frobenius scalar. It gives a constant bound for suitably spread-out source row energy, and isolates a bounded-dimensional exceptional temporal eigenspace in general. The hard case

is a source with one left row, $B = uv^T$, where v still has all mk temporal coordinates: one source row does not reduce retained rank k to one. Padding that preserves $B^T B$ does not change its spectrum; artificially spreading the energy can cost a factor k .

An ideal Gaussian regression of the exceptional residual source would pay the remaining trace. It is not an available decoder: the desired source is a component of $A - P_S A$, while sketches of A also contain the unknown in-space part. The manuscript does not claim that this contaminating component can be subtracted with the available information. This is an access-model obstruction to that proposed repair.

1.6.2 Past output, the Haar fourth moment, and symmetric dilation

The marked Bellman identity must contain both already emitted output and the future value. A terminal source mark reweights old coefficients as well as future ones. The exact telescope therefore has local self terms and mixed terms involving the odd fresh-frame kernel. Improving the self-mark constant to $8.257804885\dots$ at width $4k$ does not pay those mixed terms (`base:con:bellman`, `base:con:selfmark`).

Nor can all Haar fourth moments be majorized by their Gaussian counterparts. In the exterior-square sector the coefficients are $2/[d(d-1)]$ and $2/d^2$, respectively; a determinant minor attains the positive ratio $d/(d-1)$. The valid marked analysis locates this defect at the current mark rather than charging it through every empty stage. That placement does not eliminate the mixed old/new-mark interaction. A summable mean-forcing estimate likewise cannot simply be added to a Bellman decomposition with which it overlaps.

Symmetric dilation replaces A by $\begin{pmatrix} 0 & A \\ A^T & 0 \end{pmatrix}$. It doubles the rank and, with a common probe, changes the joint law from the two independent-stream model. Temporary diagonal-copy coefficients must remain in the state. In `base:con:sym-cone`, the adapted state bound leaves an output term of the form

$$\text{tr}[L(PP^T - P^T P)],$$

which has no fixed sign for nonscalar L . A paid two-leg expression $\|B^T K B\|_F$ does not dominate the required one-leg expression $\|B^T K A\|_F$ without further hypotheses. Scalar child weights make the commutator vanish and yield an adapted bound, but do not authorize substitution of an anticipative source. Even the dilation of an original rank-one problem has rank two, so it is not automatically the scalar case.

1.6.3 Refresh and observability must respect transported directions

Matrix storage closes several restricted geometries: matched scalar leakage, bounded exceptional levels, bounded runs between balanced resets, and specified crossed noncommuting projector families. These are sufficient geometric conditions, not properties of every HSS learner. Their observability condition must include transport. For example, $A_0 = e_1 e_2^T$, $A_1 = e_2 e_1^T$ can have $L_0 + L_1 = I$ while $L_0 + A_0 L_1 A_0^T = L_0$ and $\|A_0 A_1\| = 1$. Positive untransported window loss does not prevent indefinite survival.

Forcing strict refresh in every channel is incompatible with exact recovery of a localized rank-one matrix $e_i e_j^T$ across opposite root halves: its required direction must be preserved with norm one at every ancestor. Protected directions are not themselves an instability. If $\Pi = \ker(I - F_h F_h^T)$, the actual source satisfies $N\Pi = 0$ pathwise, and the manuscript bounds the corresponding root response. The correct issue is source excitation of nearly protected directions, not an unconditional requirement to contract every direction (`base:new:protected`, `base:new:near`).

1.7 Predictable feedback: false local charges and the later repair

Several plausible predictable-energy shortcuts are disproved even on reachable refit states. In `centered:sec:obstruction`, a rank-one state with $g = e_1$, $w = e_2$, $Y = e_3$, $Z = 0$, a row identity completion and a column quarter-turn has unchanged quotient energy, yet the current root Doob increment has variance $1/(4d) > 0$. Therefore current predictable variance cannot always be charged to current quotient dissipation. Future reserves can still pay it, so this is not a global risk counterexample.

The stronger positive-majorant obstruction in `ch:rowmajorant` concerns replacing the exact quotient J_Y by a larger positive row energy H_Y . Repeat the scalar swap/identity completions

$$Q_U = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad Q_V = I.$$

The deterministic feedback weights give unit weight to each adjacent transpose innovation, while its actual transpose output weight vanishes if another swap precedes the root. On the reachable canonical trajectory, $Y = 0$ throughout and hence $J_Y = 0$, but $H_Y = C^2$ is positive. Its scalar recursion tends to a nonzero limit. After the manuscript's three-step warm-up with one fixed nonzero source block, the source budget is one and

$$\overline{\mathcal{P}}_{G,\text{swap}} = \frac{n}{16d^2q} + O(1), \quad \mathcal{P}_{G,\text{swap}} = 0.$$

At $q = 4$ the spurious slope is $1/256$. Increasing to any other fixed width does not remove the linear growth of this upper bound. The lost terms in $J_Y = H_Y - D_Y D_Y^T - \Delta \Delta^T$ are needed for cancellation.

The earlier exact Doob decomposition, passive mean filter, five future-variance registers, and bound on the marked state at each deterministic time remain valid. A current-state estimate is not an estimate of a maximum or a sum over times. Coordinate-averaged response bounds leave a positive anisotropic remainder unless the actual bases have the additional symmetry. Backward weights are not automatically monotone either. These were missing premises in the earlier proof route, not evidence that the final decoder fails.

Repaired status. Chapters `ch:feedback`–`ch:uniform` retain the future projectors under legitimate frozen-stream conditioning, route complementary loss, and preserve the signed retained-coefficient combination. In the notation of that proof, the cancellation variable $R_i = f_i h_i^T + M_i^T$ has a transport recursion with the otherwise duplicated input terms canceled. Deterministic frame transport then bounds the needed output, including coefficients already emitted. The resulting fixed-space theorem has factor 1607 with $8k$ fresh refit products; larger fixed widths yield sharper constants and the stated near-optimal family.

Thus the historical logarithmic oversampling, exponential initial refit bound, and unpaid predictable interaction are superseded as obstacles to fixed-space refitting. The counterexamples to their discarded intermediate inequalities remain correct and explain why the repaired proof keeps signed and noncommuting information. None of these repairs establishes the quality of a noisy learned space.

1.8 Adaptive annihilation: generic erasure and actual inverse-growth families

The later chapters of `hss_complete_proof_chain.tex` replace unrestricted adaptive examples by specified learners and fixed input matrices. The distinctions between their conclusions

are important: an erased sample, an exploding unweighted inverse, and a positive physical approximation loss are three different outcomes.

1.8.1 Perfect conditioning and comparator containment do not suffice

The generic erasure construction in `ch:basisobs` fixes $A = ue_j^T$, where $u = (u_a + u_b)/\sqrt{2}$ is supported in two children and j is external. On a positive-probability event, the Gaussian child maps have enough physical nullspace to choose unit vectors v_a, v_b satisfying

$$\Omega_a^T v_a = a_0, \quad \Omega_b^T v_b = b_0,$$

where a_0 is the normalized exterior row $e_j^T \Omega$ and b_0 is perpendicular to it. The compressed design is a row isometry. Nevertheless its row space contains the exterior anchor, so nullification erases the outgoing rank-one observation. The row templates can contain the comparator exactly, and all these choices can be nested.

This disproves an inference from containment and good conditioning alone. The selected column directions are admissible adaptive choices, not claimed to be the actual canonical rule's trajectory. The same chapter's path example shows a different limitation: a fixed HSS-rank-one matrix can introduce a new cut direction at every ancestor while the old direction restricts to zero in the next child. The rank- k version has kh such innovations along a path. It excludes an argument that assumes only k directions ever occur along a path, but gives no kh query lower bound because different levels and nodes may be probed simultaneously.

1.8.2 An actual phantom-direction family

Theorem `actual:thm:inverse` concerns the canonical un-nullified commutator learner. Fix $B = e_j v^T$, where v is a unit vector supported in the opposite root half and is nonzero on every coarse leaf. On that half the true outgoing block is zero, but the commutator contains $-\zeta(v_I^T \Omega_I)$, with $\zeta = \Psi^T e_j$. The full raw lift retains a direction related to $(G^\dagger)^T \zeta$. This unused, or phantom, direction attracts the query images of both children toward the same exterior anchor.

For retained width $r > 1$, leaf width $m = 2r$, and $q > 2r$, put

$$\mu = \frac{q - m}{m - 2}, \quad \beta = \frac{q - 2r + 1}{q - r} < 1.$$

The exact graph-angle recursion and a Gaussian norm estimate imply, at a parent whose children have height h and $m2^h \geq \max(q, 8)$,

$$\Pr\{\sigma_{\min}(G^{(h+1)})^2 \leq 36m\mu\beta^h\} \geq \frac{1}{2}, \quad \mathbb{E}\|(G^{(h+1)})^\dagger\|_{\text{op}}^2 \geq \frac{\beta^{-h}}{72m\mu}.$$

The designs remain full row rank almost surely. At $r = 4, m = 8, q = 16$, the lower bound is $(4/3)^h/768$. Increasing to another fixed width ratio slows the rate but does not restore a depth-uniform full inverse moment.

The matrix is still represented exactly. The collapsing orientation has zero true outgoing comparator, and the nonzero row and column ranges remain contained. Thus this is an actual-rule obstruction to an *unweighted inverse intermediate bound*, not a lower bound on approximation error.

1.8.3 A saturated symmetric family removes the inactive-cut explanation

Theorem `ports:thm:inverse` uses

$$B = \begin{pmatrix} 0 & UU^T \\ UU^T & 0 \end{pmatrix},$$

where U has k orthonormal columns and rank- k restrictions on every leaf; repeated, appropriately normalized identity blocks give an explicit choice. Every relevant incoming and outgoing cut has rank k . For $r \geq 2k$ and $q \geq 2r + k$, the raw commutator has rank at most $2k$, so the actual full-range branch retains both the signal and its external query port. The children's query images again align around that common port.

With

$$t_0 = \frac{k(q-2r)}{2r-k-1}, \quad \lambda = \frac{q-2r+k}{q-r} < 1,$$

the exact angle recursion yields, at a specified height- j node,

$$\mathbb{E}\|G_I^\dagger\|_{\text{op}}^2 \geq \frac{\lambda^{-(j-1)}}{8rq t_0}.$$

At $r = 4k, q = 16k$ this is $(7k-1)(4/3)^{j-1}/(4096k^4)$. The proof uses Cauchy-Schwarz and Jensen on the joint angle and physical-probe norm; it does not factor them as independent. Expectations can be extended nonnegative expectations; finiteness at each height is not part of the conclusion.

Exact representation still holds. In particular the error projector annihilates the lifted external port:

$$(I - U_I^{\text{loc}}(U_I^{\text{loc}})^T)(G_I^\dagger)^T H = 0.$$

Even saturation and nonzero cut signals therefore do not justify a uniform full inverse bound. The correct error readout may cancel the very directions responsible for inverse growth.

1.9 Inherited error and the actual noisy annihilator failure

1.9.1 A clean parent can create new loss with well-conditioned designs

Theorem `return:thm:example` follows the actual hybrid rule at $k = 1, r = 2, m = 4, q = 16, N = 32$. For adjacent leaves a, b and an external coordinate j , set

$$B = (e_{a,1} + e_{b,1})e_j^T, \quad E[a, b] = e_3e_1^T + 2e_2e_2^T + \tfrac{3}{2}e_4e_3^T.$$

All other blocks of E vanish. Thus E is internal to the parent $I = a \cup b$, and its current cut commutator is identically zero. On explicitly specified open probe events the current row design has singular values in $[3/4, 5/4]$, the column design is close to a row isometry, and the surviving external anchor is bounded away from zero. The current raw sample tails vanish in both orientations.

Nevertheless, for $A_\varepsilon = B + \varepsilon E$, the inherited and newly discarded row losses at the reference probes are

$$h_U^2 = \frac{\varepsilon^2}{1 + \varepsilon^2}, \quad e_U^2 = \frac{\varepsilon^2 M^2 (1 + \varepsilon)^2}{((1 + \varepsilon)^2 + \varepsilon^2)^2 + \varepsilon^2 M^2 (1 + \varepsilon)^2}.$$

Here M is arbitrary and $0 < \varepsilon \leq \min(0.1, 0.1/M)$. Consequently $e_U^2/h_U^2 \rightarrow M^2$ as $\varepsilon \downarrow 0$. Strict local singular-value gaps make the inequalities persist on positive-probability neighborhoods.

This excludes charging new physical loss only to the current crossing residual or current truncation tails. It also excludes a uniform pointwise inherited-loss gain under those current-design safeguards. The event probabilities need not be uniform in M , so the result is compatible with the manuscript's unconditional mean estimates in the narrower clean model.

The failure occurs despite a valid all-depth theorem for the unlifted data. If $E = A - B$ for any fixed HSS-rank- k comparator, Theorem `actual:thm:datafit` proves for the actual hybrid selections, including singular internal designs,

$$\sum_I \mathbb{E}(\|\mathcal{R}_{U,I}\|_F^2 + \|\mathcal{R}_{V,I}\|_F^2) \leq 6q^2 L \|E\|_F^2.$$

The proof uses only that each projected comparator commutator has rank at most $2k$ and that the selected query space minimizes the corresponding rank- $2k$ tail. Its first physical leaf layer also has a residual-relative bound. Neither result converts a small query-space residual into a small physical loss under an adaptive opposite annihilator.

There is an additional useful distinction between a vanishing *local* tail and vanishing tails throughout the tree. Theorem `return:thm:zero` proves, for each fixed A and the stated hierarchical branch, that the total tail is zero almost surely exactly when A belongs to the HSS-rank- $2k$ class; the learned space then contains A . The probability-one event may depend on A and is not simultaneous over all sketch-dependent matrices. This qualitative endpoint is compatible with arbitrarily poor quantitative behavior near it and with the local clean-parent example.

1.9.2 A fixed-input family disproves the annihilate-first basis bound

Theorem `anchor:thm:main` is stronger: it disproves the desired basis-quality bound for the specified unchanged annihilate-first hybrid. Fix $r \geq 2$, $q > 4r$, and a depth parameter h ; put $n = 4r2^h$, $N = 4n$. Let I and K be adjacent blocks of size n , $P = I \cup K$, and j the first coordinate outside P . Set $u = n^{-1/2}\mathbf{1}_I$ and $B_h = ue_j^T$. Choose two unit Frobenius-norm perturbations with mutually disjoint supports:

$$E_0 = \frac{1}{\sqrt{n}} \{\text{permutation exchanging adjacent } 2r\text{-leaf blocks in } I\},$$

$$E_1 = \frac{1}{\sqrt{r}} \sum_{a=1}^r e_{n+a-1} e_{j+a}^T.$$

The fixed input is $A_h = B_h + \delta_h E_0 + \varepsilon_h E_1$. The comparator is HSS rank one and $e_k(A_h) \leq \delta_h^2 + \varepsilon_h^2$.

The arbitrarily small positive perturbation $\delta_h E_0$ activates otherwise unused column directions. Across the clean subtree these directions trap the row anchor $g = e_j^T \Omega$ in the learned column query spaces. The original probe reuse is essential to this accumulation. With

$$\alpha = \frac{q - 2r + 1}{2(q - r)} < \frac{1}{2}, \quad \mu = \frac{q - 2r}{2r - 2},$$

the expected squared surviving anchor is at most $q\mu\alpha^h$. The sibling perturbation $\varepsilon_h E_1$, with $\varepsilon_h = \alpha^{h/4}$, then displaces the almost annihilated signal in the rank- r truncation. At $\delta = 0$ the canonical zero-sample branch is different; exact recovery on that stratum gives no continuity through positive δ .

For each fixed h , bounded convergence permits a *deterministic* choice $0 < \delta_h \leq \varepsilon_h^2$ along a predetermined sequence. It is chosen by an expectation criterion, never from the realized probes. In the maintained parameters $r = 2k$, $q = 16k$, the theorem is

$$\mathbb{E} d_{S_{\text{hyb}}(A_h)}(A_h)^2 \longrightarrow 1, \quad L_h e_k(A_h) \longrightarrow 0, \quad L_h = h + 4.$$

The optimum is strictly positive for every $\delta_h > 0$, since an interior sibling block has rank $2r > k$. Therefore no finite constant C_b can make $\mathbb{E}d_{\mathcal{S}_{\text{hyb}}}(A)^2 \leq C_b Le_k(A)$ hold for all fixed inputs.

The first adaptive-merge theorem is not contradicted. There the child templates have a stream-separation property. Here a zero-comparator orientation can store harmful query information at zero immediate comparator cost; at later levels it acts through the opposite annihilator, where stream separation is absent.

1.9.3 What the counterexample says about fallbacks and unions

Normalized direct sums of T aligned copies preserve HSS rank one while making the comparator have global rank T with equal singular values $T^{-1/2}$. The expected hybrid-space loss still tends to one. For $T > 4k$, any global-rank- $4k$ candidate has limiting squared error at least $1 - 4k/T$, and a physical-leaf-diagonal candidate misses the entire unit comparator. Hence selection among these fallbacks before further nonlinear HSS projection does not repair the example (`anchor:cor:directsum`, `anchor:eq:globalfallback`). No assertion is made about arbitrary subsequent best-HSS projection, which can change the bases.

The un-nullified companion retains the displaced signal on this family. The original manuscript proves exact two-view rescue and then a mean-square small-noise result in `ch:tangent` and `ch:nonlinear`. These are positive results for the specified family and interfaces. The two complete hierarchies are learned separately before their physical union is taken. They are not a recursively union-fed learner, and they can have different child templates. A proof about a distinguished clean cut is not a whole-space, all-input basis theorem.

In particular, the two-view hyperplane theorem conditions on a full-column-rank lift, a projector, and a nonzero anchor in both projector sectors, while leaving an independent Gaussian competitor block. Its projector-complement rank must be at least $r + 1$. The two leading- r spaces then span the full $(r + 1)$ -dimensional lift almost surely (`tangent:thm:twoview`). Learned opposite designs on an arbitrary input cannot simply be declared independent to fit this theorem.

The tangent result first fixes height and lets the perturbation vanish. The nonlinear moment argument supplies the stated expectation interchange in that local model. It does not prove a height-uniform finite-noise remainder. Changing the order of the height and perturbation limits, or importing the same subtree locality to general coupled HSS cuts, would be an additional premise.

1.10 Rank slack and simultaneous interpolation do not resolve the interface

For $S = C + F$ with $\text{rank } C \leq k$, the leading- r projector P satisfies the valid local spectral estimate

$$\|(I - P)S\|_{\text{op}}^2 \leq \frac{\|F\|_F^2}{r - k + 1}.$$

Consequently, if $\text{rank } Z \leq k$ and $\|Z\|_{\text{op}} \leq 1$, then $\|(I - P)SZ\|_F^2 \leq k\|F\|_F^2/(r - k + 1)$. The target here is the observed sum S , not its comparator part C . Taking $S = 0$, $F = -C$, with C perpendicular to the canonical zero-sample completion, prevents replacing S by C with a universal contraction coefficient below one (`ch:slack`).

Likewise, capturing both local innovations does not necessarily produce a matrix interpolating both complete sketches. In `ch:bicriteria`, the two-sided construction can annihilate the forward residual yet leave a transpose term

$$(I - Q)(B_Z^T - B_Y^T)(I - P)\Psi.$$

The underlying generic dimension obstruction is explicit. At width $q = 3r$, the two-sketch measurement rank is $6Nr - 9r^2$, whereas the saturated HSS-rank- r parameter count in the stated tree model is $5Nr - 6r^2$; the former is larger for $N > 3r$. Generic observations therefore have no exact HSS-rank- r interpolant. This rules out an exact interpolation step in that argument, not an approximate HSS recovery theorem.

Finally, adding a large physical-leaf-diagonal matrix does not regularize a raw learning trajectory: on a full-rank trajectory, $D + \varepsilon E$ has the same learned bases as E , and the homogeneous reconstruction is $D + \varepsilon \hat{E}$. Thus closeness to an exactly representable leaf-diagonal input is not, by itself, a stability proof (`ch:fallback`).

1.11 Local equivariance is insufficient; guards leave a mandatory cost

1.11.1 A broad local-equivariant model has genuine physical growth

Chapter `ch:budget` fixes leaf sources before one Gaussian probe and permits subtree-local, query-rotation-equivariant selections. It proves the exact innovation and energy identities before constructing a bad selection. Write

$$\lambda = \frac{q - 2r}{q - r}, \quad \kappa = \frac{q - r - 1}{q - 2r - 1}, \quad \rho_0 = \frac{r}{q - r - 1}.$$

For child height t , the normalized innovation energy is $m_I = \lambda^t \beta_I$. If V_I is the retained inverse trace and a_I its innovation-weighted mean, the merge cost is

$$\mathbb{E} \|z_I\|_F^2 = (\kappa - 1)(a_a + a_b) + \kappa(m_a v_b + m_b v_a), \quad v_i = \mathbb{E} V_i.$$

The cross-child terms already show why innovation decay alone does not control physical error.

In Theorem `budget:thm:counter`, one root subtree has a single unit source and fixed coordinate selections; the other has zero source but retains the weakest r Gram directions at every merge. Its errors vanish, while its inverse mean grows. At the root an admissible selection discards the entire fitted coefficient vector. All designs are full row rank and all second moments are finite at each finite height, but

$$\mathbb{E} \|E_{\text{root}}^{\text{out}}\|_F^2 \geq 1 + \rho_0 + \kappa \rho_0 (\lambda \kappa)^h, \quad \lambda \kappa = 1 + \frac{r}{(q - r)(q - 2r - 1)} > 1.$$

At $r = 2, q = 16$ the lower bound is $15/13 + (2/11)(78/77)^h$. For $r = 1$ the sharper sphere-angle calculation replaces κ by

$$\chi_q = \frac{q - 2 + 2C_q}{q - 3}, \quad C_q = \frac{\Gamma(q/2)}{\sqrt{\pi} \Gamma((q - 1)/2)},$$

and gives $\lambda \chi_8 > 1.3778$.

This disproves a theorem uniform over the entire local-equivariant selection class. The construction deliberately chooses weak directions and, at the root, discards the fitted target. It is not the canonical full-fit rule or a fixed-matrix counterexample to the maintained HSS union. Its purpose is to show that locality and rotational symmetry do not replace analysis of the actual selection.

1.11.2 A sufficient relative guard can conflict with exact containment

The same chapter proves a positive conditional theorem under $V_I \leq \gamma \min(V_a, V_b)$ at ordinary nodes. The physical bound contains $\sum_{t < H} (\gamma\lambda)^t$; it is uniform in height when $\gamma\lambda < 1$ and linear at equality. Choosing the leading Gram directions supplies $\gamma = 1$, but can discard a required signal. Stability of this selection therefore does not by itself define a valid HSS learner.

Proposition `budget:prop:infeasible` makes the conflict quantitative. Let

$$K = \begin{pmatrix} I_r & (1 - \varepsilon)e_1e_1^T \\ (1 - \varepsilon)e_1e_1^T & I_r \end{pmatrix}, \quad Q = \frac{1}{\sqrt{2}} \begin{pmatrix} e_1 \\ -e_1 \end{pmatrix}.$$

Both child inverse traces equal r , but the mandatory direction Q has Gram eigenvalue ε . Among all rank- r compressions containing it,

$$V_{\text{opt}} = \frac{1}{\varepsilon} + \frac{1}{2 - \varepsilon} + r - 2.$$

No fixed finite relative guard can preserve every prescribed signal on every positive-definite realization. This is a local algebraic obstruction; it says nothing about the frequency or source weight of the bad configuration under a specific learner.

1.11.3 Optimal padding isolates, but does not remove, the core obligation

The exact optimizer in `budget:thm:completion` retains an orthonormal mandatory frame Q of rank $s \leq r$. In coordinates $[Q, Z]$, write

$$K = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix}, \quad \mathcal{S} = D - B^T A^{-1} B, \quad \mathcal{M} = I + B^T A^{-2} B, \quad \phi = \text{tr}(A^{-1}).$$

If ν_j are the increasing eigenvalues of $\mathcal{S}^{-1/2} \mathcal{M} \mathcal{S}^{-1/2}$, block inversion and generalized Rayleigh–Ritz give $V_{\text{opt}} = \phi + \sum_{j=1}^{r-s} \nu_j$. This can be computed locally without new matrix products. The later additive theorem `core:thm:additive` sharpens the bookkeeping to

$$V_{\text{opt}} \leq \phi + \sum_{i=s+1}^r \lambda_i(K)^{-1} \leq \phi + \min(V_a, V_b),$$

with the eigenvalues of K in decreasing order. The unavoidable quantity is the mandatory-core inverse ϕ , not optional padding. A full-rank truncated sample has $s = r$ and no optional freedom at all.

The fourth-moment result `core:thm:fourth` is valid under the local model even for unstable admissible selections:

$$\mathbb{E} \|w_I\|_F^4 \leq d_0(d_0 + 2)\chi^t \beta_I^2, \quad d_0 = q - r, \quad d = q - 2r, \quad \chi = \frac{d(d+2)}{d_0(d_0+2)}.$$

Thus a uniform marginal L^2 bound on ϕ pays its dependent product with the innovation by Cauchy–Schwarz. At $r = 2, q = 16$, the decay is $\sqrt{\chi} = \sqrt{3}/2$, not $\lambda = 6/7$. The additive inverse-budget theorem allows inverse means to grow linearly while the physical cost remains summable.

For a *fixed physical* nested core, the core projection has the required Gaussian marginal law even if its optional completion is adaptive. The original chapter obtains explicit uniform constants for this restricted setting. A noisy selected core need not be fixed; replacing it by an unknown comparator changes the learner and invalidates that law. General cuts can also share exterior probe rows and use unequal opposite templates, violating the independent-subtree interface. These are the remaining premises identified in `core:sec:pilot` and `core:sec:checks`, not counterexamples to every possible guarded algorithm.

1.12 Coverage of the historical ledger and scope of the remaining premise

The following map accounts for all 66 numbered routes in the original `base:new:route-history`. Numbers are grouped by mathematical issue rather than by the order in which they were investigated. Positive restricted theorems appear because their hypotheses delimit the corresponding obstruction.

Original routes	Taxonomy and precise qualification
1, 2, 7	Section 1.1.1: exact recovery and the first two levels do not imply all-depth reuse; freshness changes query cost.
5, 6, 10	Section 1.2: adaptive inverse moments, Stein correction, and legitimate conditional Wishart laws.
8, 9, 12, 55, 56	Section 1.3: tails, validation, fitting injectivity, saturated identification, and local versus uniform conditioning.
3, 4, 14, 15, 17, 30	Section 1.4: inactive reconstruction, common coefficients, signed merges, correlated feedback, and scalar storage.
13, 16, 18	Sections 1.5 and 1.12: finite-profile hypotheses, restricted coordinate paths, and the loss in absolute overlap bounds.
19–29, 36–38, 45, 46	Section 1.5: source orthogonality, full covariance state, conditional amplification, temporal marks, embedding, near-protection, moment envelopes, and repaired restart/source accounting.
31–35, 39–44, 47–54, 57, 58	Section 1.6: directional storage, transported refresh, covariance metrics, coherent rows, inaccessible auxiliary data, Bellman marks, Haar defect, dilation, and symmetry.
11, 59–66	Section 1.7: initial refit bounds, passive filter, Doob product, quotient/variance interfaces, current-state control, and their later fixed-space repair.

For completeness, the finite-profile transfer theorem called F^+ in `base:new:fp` assumes a specified centered rank-one conditional response profile. Isotropy of an HSS sketch does not imply that profile. The factor-10 coordinate-path theorem exploits a nonbranching coordinate geometry. Neither is a substitute for proving the hypotheses along a general adaptive trajectory. The source-history ledger’s finite temporal certificates and repeated-eigenvalue symmetry arguments have the same restriction: their finite family or actual symmetry must be present.

Several ideas were considered but never promoted to completed algorithms: repeated independent residual corrections, redundant retained dimensions intended to balance arbitrary bases, query-coordinate gauge randomization, and a direct whole-space small-ball estimate. The record contains no theorem showing that one of them simultaneously preserves exactness on the original rank class, costs $O(k)$ products, and has a uniform residual-relative bound. Their absence is a missing construction, not an impossibility theorem.

The later obstruction families, not contained in the first 66-route chronology, are covered explicitly above: independent half solves, adaptive erasure and path novelty, tree-nuclear selection, rank-slack and interpolation limitations, both actual inverse-growth families, the well-conditioned inherited-return example, the actual annihilator failure and its fallbacks, the broad local-equivariant counterexample, and mandatory-core guard infeasibility. All statements in this chapter concern the rules, models, and probability quantifiers stated with them.

The original manuscript therefore ends with a sharply separated dependency chain. Independent refitting of a fixed nested space is proved. Exact offline rank reduction is a valid query-only step. The specified annihilate-first learner’s general basis bound is false. The maintained two-hierarchy union has exact and small-noise results on the stated special family, but its all-input basis-quality estimate is not supplied by those results or by the abstract local regression the-

orems. This is the original manuscript's remaining interface, not a lower bound against all $O(k)$ -query HSS algorithms.

Verification scope. The exact fractions quoted above are historical finite certificates preserved in the original manuscript. No Monte Carlo experiment or historical certificate engine was rerun for this consolidation. A finite simulation can check an identity or illustrate a specified trajectory; it does not establish an all-depth probabilistic estimate. All rank, inverse, and nullspace statements here concern exact arithmetic with the displayed conditioning, rather than a floating-point rank threshold.

Obstructions and limits established after the original manuscript

This chapter records what the subsequent investigations ruled out, what they repaired, and what remained unresolved when research stopped. Its scope is deliberately narrower than an impossibility theorem for HSS approximation. A counterexample to a selector, an adaptive moment estimate, or one prescribed hierarchy is not a lower bound against every algorithm using a depth-independent number of queries. The general conjecture—an $O(k)$ -query algorithm with expected squared Frobenius error $O(L \text{OPT}_k(A))$ and final HSS rank at most k —was not proved or disproved.

All matrices in the probabilistic counterexamples below are fixed before the Gaussian probes. Some competing interpolants, conditional states, or event windows are subsequently selected from the observations; those quantifiers are stated explicitly. The original manuscript’s adaptive nullification and annihilate-first counterexamples are recalled only where needed to distinguish later learners. The detailed follow-up catalogue is based on the source notes and their recorded audits, not on the numerical summaries alone. `hss_audit_report.md` provides the chronological overview.

2.1 Global interpolation and minimum-norm selectors

2.1.1 A fixed rank-one input with smaller rank-one HSS interpolants

The global-selector extension removes the qualification that a regularizer obstruction requires nominal rank at least two. Let the two root children I, J each have size n , let $q < n$, and draw independent standard Gaussian probes $\Omega, \Psi \in \mathbb{R}^{2n \times q}$. For fixed unit vectors $u, v \in \mathbb{R}^n$, set

$$B = \begin{bmatrix} 0 & uv^T \\ 0 & 0 \end{bmatrix}.$$

This fixed input is HSS rank one on the complete coordinate tree and has unit Frobenius norm. Almost surely the systems

$$\Psi_J^T a = \Psi_I^T u, \quad \Omega_I^T b = \Omega_J^T v$$

have solutions; choose their minimum-Euclidean-norm solutions. For every real t , the observed-data-dependent competitor

$$C(t) = \begin{bmatrix} (1-t)ub^T & tuv^T \\ -(1-t)ab^T & (1-t)av^T \end{bmatrix}$$

satisfies

$$C(t)\Omega = B\Omega, \quad C(t)^T\Psi = B^T\Psi.$$

It remains HSS rank one: at any node contained in I , outgoing and incoming exterior blocks have the respective common factors u_S and b_S^T ; at a node contained in J , the corresponding factors are a_S and v_S^T . Thus the assertion concerns every cut, not merely the root split.

Writing

$$\eta = \|a\|^2 + \|b\|^2 + \|a\|^2\|b\|^2,$$

the disjoint supports of the four blocks give

$$\|C(t)\|_F^2 = t^2 + (1-t)^2\eta.$$

Its minimum occurs at $t_* = \eta/(1+\eta)$, and

$$\|C(t_*)\|_F^2 = \frac{\eta}{1+\eta} < 1, \quad \|B - C(t_*)\|_F^2 = \frac{1}{1+\eta}.$$

Consequently minimum-Frobenius-norm HSS interpolation, including minimum norm as a tie rule for exact empirical least squares, fails almost surely on this fixed exact input. A universal relative guarantee would require zero error because $\text{OPT}_k(B) = 0$ for every $k \geq 1$.

The exact displayed error belongs to the constructed competitor, not to an arbitrary optimizer. The latter nevertheless has a quantitative risk lower bound. If $n > q + 1$, put

$$\rho = \frac{q}{n - q - 1}, \quad \mu = 2\rho + \rho^2.$$

Inverse-Gram averaging gives $\mathbb{E}\|a\|^2 = \mathbb{E}\|b\|^2 = \rho$; the two vectors use independent probe streams, so $\mathbb{E}\eta = \mu$. Concavity of $x/(1+x)$ and the reverse triangle inequality in the space of square-integrable matrices imply, for every measurable minimum-norm selection $\hat{C}$,

$$\mathbb{E}\|B - \hat{C}\|_F^2 \geq \left(1 - \sqrt{\frac{\mu}{1+\mu}}\right)^2.$$

This tends to one as $n/q \rightarrow \infty$. Existence of the minimizer follows from closed rank constraints and compact norm sublevel sets. These statements are established in `hss_global_selector_observation.tex`.

2.1.2 Tree penalties and exact leaf-block calibration

For constant u, v on balanced root halves, the outgoing-plus-incoming cut nuclear norm of B at a depth with m nodes per half is $2\sqrt{m}$. All cuts of $C(t)$ have rank at most one. The partition bound

$$\sum_S (\|C[S, S^c]\|_F^2 + \|C[S^c, S]\|_F^2) \leq 2\|C\|_F^2$$

and Cauchy–Schwarz bound its nuclear sum by $2\sqrt{2m}\|C\|_F$. On $\eta < 1$, the smaller interpolant has $\|C(t_*)\|_F < 1/\sqrt{2}$ and improves every depth strictly. It therefore improves every nonzero nonnegative deterministic depth-weighted sum of these cut nuclear norms. Its ordinary matrix rank is at most two, so its ordinary nuclear norm is also smaller than that of B on this event. Markov’s inequality gives probability at least $\max(0, 1 - \mu)$ for the event. The same strict comparison defeats positive penalization because both empirical residuals vanish.

Supplying all physical-leaf diagonal blocks exactly does not repair these selectors. Mark one leaf in each sibling pair active and the other inactive. Support u, v on the active leaves of their respective halves, and restrict b, a to inactive leaves. The free dimension in each interpolation system is then $d = n/2$; assume $d \geq q$. On every physical leaf K , at least one factor of the corresponding diagonal outer product vanishes, so

$$C(t)[K, K] = B[K, K] = 0$$

exactly. The smaller-norm construction is otherwise unchanged.

For $d > q + 1$, replace ρ, μ by $\rho_d = q/(d - q - 1)$ and $\mu_d = 2\rho_d + \rho_d^2$. The minimum-norm lower bound remains valid. With u, v spread uniformly over active coordinates, the true leaf-depth nuclear sum is $2\sqrt{m/2}$, while the competitor still obeys $2\sqrt{2m}\|C\|_F$. The stronger event $\eta < 1/3$, of probability at least $\max(0, 1 - 3\mu_d)$, gives $\|C(t_*)\|_F < 1/2$ and improves every depth, including the leaves. Any finite penalty depending only on the supplied leaf diagonal blocks takes the same value on both candidates and cannot reverse this comparison. A single-active-leaf version also gives calibrated minimum-norm failure, with free dimension $n - s$ for leaf size s . See `selector_with_leaf_data.md` and the final theorem in `hss_global_selector_obstruction.tex`.

The alternatives above depend on the probes. They disprove specified global selection principles; they do not say that two fixed inputs are indistinguishable with positive probability, or that a different structured decoder cannot recover each fixed HSS input exactly.

2.1.3 Certificates restore exact uniqueness, not robust selection

The trajectory-certified selector imposes injectivity of each complete physical template together with its exterior cut anchor. Under these certificates, equal transcripts determine equal outgoing and incoming intersections, hence equal canonically completed child frames. Induction then gives a common coefficient space; successive two-sided compression, ending with an injective root core, proves exact uniqueness. The false interpolants above violate these full-template certificates, including the endpoint where a minimal cut anchor alone would not detect them.

For a fixed exact input the comparison templates are deterministic, so each $q \times p_I$ certificate matrix is Gaussian and $\mathbb{E}\sigma_{\min}^{-2} \leq p_I/(q - p_I - 1)$. This marginal bound cannot be replaced by a tree-size-independent bound on the maximum certificate cost: disjoint leaves provide arbitrarily many independent Gaussian direction images. Hard thresholds also exclude some exact inputs with positive probability. The quantitative compression inequality gives products of levelwise factors, rather than the desired robust residual-relative bound for changing candidate spaces. The exact measurable selector is therefore a valid positive endpoint, not a completed approximation algorithm. See `certified_selector_route.md` and `certified_selector_audit.md`.

2.2 The one-sided Gaussian pilot obstruction

Removing transpose data from a pilot would make its row spaces independent of a later transpose stream, but exact bounded-rank row containment already fails for that restricted acquisition model. Fix query width q and output rank r . Take powers of two $d \geq 2r$ and $M \geq 2q + 2$. Partition one root half I into d aligned subtrees K_i of size M , choose fixed unit coordinates $e_i \in K_i$, and choose a fixed unit coordinate v in the other half. For arbitrary t_i and b_i supported in K_i , define

$$u = \sum_{i=1}^d t_i e_i, \quad A = \sum_{i=1}^d e_i b_i^T + uv^T.$$

Every such matrix is HSS rank one, including at cuts inside the K_i . The all-cut verification uses the single row factor e_i inside a block and the single exterior factor v above unions of complete blocks; no large unrestricted leaf is hidden in the construction.

A pilot observing only $(\Omega, A\Omega)$ obtains, apart from zero rows, exactly

$$y_i = X_i^T b_i + t_i \gamma, \quad X_i = \Omega_{K_i}, \quad \gamma = \Omega_J^T v.$$

To obtain a fixed-input impossibility, use an input prior only in the proof: independently take $t_i \sim N(0, 1)$ and $b_i \sim N(0, \tau^2 I_M)$ before drawing Ω . Conditional on the complete observations, the posterior covariance of t is diagonal with strictly positive entries

$$w_i = \left[1 + \gamma^T (\tau^2 X_i^T X_i)^{-1} \gamma\right]^{-1}.$$

Thus u has a posterior density on the entire fixed d -dimensional span of the e_i . After also fixing any independent algorithmic seed, a returned space of dimension at most $r < d$ contains u with posterior probability zero. Fubini converts this statement into failure for almost every fixed input under the proof prior, with the bad-input set allowed to depend on the pilot.

There is also a quantitative Bayes loss bound. A rank- r projector captures at most r from a covariance bounded by the identity, giving

$$\mathbb{E} \|(I - P_S)u\|^2 \geq \frac{d}{1 + q/[\tau^2(M - q - 1)]} - r.$$

For $\tau = 2$ and the displayed dimensions this is at least $3d/10$. The result permits unlimited computation and disclosure of the entire parametric family. It does not cover structured forward queries: on this very family the single query $Av = u$ reveals the desired vector. Transpose products and independently trained two-sided pilots are also outside its scope. The theorem and independent review are `one_sided_gaussian_pilot_obstruction.md` and `one_sided_gaussian_pilot_audit.md`.

2.3 Why the screened-source moment proposed by Claude is false

2.3.1 The valid deterministic cancellation

For a full-row-rank design $G = U\Sigma V^T$, lift $L = (G^\dagger)^T$, response J , and retained projector P with $PLJ = LJ$, split the singular directions into the weakest d and the remaining directions. If $J_1 = V_1^T J$ has full row rank, then

$$(I - P)L = (I - P)U_2\Sigma_2^{-1}[V_2^T - J_2J_1^\dagger V_1^T].$$

This algebraic identity is correct. The failed proposal was to discard the remaining left projector and assert a uniformly controlled second moment for the resulting screened source. Weak singular values must indeed be the removed block; sorting in decreasing order would not even give the claimed deterministic inverse bound. Full rank of G also does not imply the required response rank of J_1 .

2.3.2 An infinite moment at an ordinary Gaussian leaf

The counterexample uses the actual full-range learner's leaf dimensions $m = 16, q = 64, s = 4, r = 8$. Fix orthonormal physical vectors a, e inside a leaf and a unit v outside it, and take

$$A = av^T, \quad B = (a - e)v^T, \quad E = ev^T.$$

The comparator is legal but need not be best. Here G is an ordinary Gaussian leaf design and, for an independent Gaussian response row $\gamma = v^T \Omega$, $J = G^T a \gamma$ and $J(E) = G^T e \gamma$. Sort the singular values increasingly, and write $c = U^T a, d = U^T e$. Screening only the weakest coordinate gives

$$Q(E)_i = \sigma_i(d_i - c_i d_1 / c_1) \gamma, \quad i \geq 2.$$

Consequently

$$\sigma_2^{-2} \|Q(E)\|_F^2 \geq \|\gamma\|^2 \left(1 + \frac{d_1^2}{c_1^2}\right).$$

Haar orientation gives a density for c_1 positive at zero and $\mathbb{E}[d_1^2 \mid c_1] = (1 - c_1^2)/(m - 1)$. The expectation is infinite, even conditional on any nonzero γ . All stated rank assumptions hold almost surely. In contrast, the actual fit is $a\gamma$, so the retained space contains a and

$$\|(I - P)LJ(E)\|_F^2 \leq \|\gamma\|^2.$$

Its expectation is at most four. The discarded left projection can cancel a nonintegrable response ratio.

2.3.3 Linear scaling even for an exact best comparator

Fix orthonormal a, e inside the leaf and orthonormal v, w outside it. For $0 < \varepsilon < 1$, let

$$A = av^T + \varepsilon ew^T, \quad B = av^T.$$

The outgoing cut has singular values $1, \varepsilon$, and B attains the cut lower bound globally. Thus $\text{OPT}_1(A) = \varepsilon^2$. Put

$$\gamma = v^T \Omega, \quad \eta = w^T \Omega, \quad g = \|\gamma\|^2, \quad \kappa = \|\eta\|^2, \quad h = \gamma \eta^T, \quad \Delta = g\kappa - h^2.$$

The same one-coordinate screening has a lower-bound integrand whose Haar average is exactly

$$\frac{\varepsilon g \kappa}{2\sqrt{\Delta}}.$$

One obtains this by setting $t = c_1/d_1$, using its standard Cauchy density, and integrating the resulting rational function. For independent Gaussian response rows in $\mathbb{R}^4$, the radial and angular moments give

$$\mathbb{E}[\sigma_2^{-2} \|Q(E)\|_F^2] \geq \frac{9}{4}\varepsilon.$$

Appending the natural comparator separator $T = \gamma^\dagger$ still gives

$$\mathbb{E}[\sigma_2^{-2} \|Q(E)T\|_F^2] \geq \frac{1}{4}\varepsilon.$$

Neither expression is uniformly $O(\varepsilon^2)$. Yet the full fit $a\gamma + \varepsilon e\eta$ has range $\text{span}(a, e)$ almost surely, so the actual filtered source, with or without T , is identically zero. This defeats the proposed proof lemma more strongly than a badly chosen comparator would, while leaving the leaf learner successful. See `claude_screened_residual_audit.md` and `screened_source_counterexample_audit.md` for the exact integrals, corrected dimensions, perturbation expansion, and review.

The same audit also rejects two unrelated shortcuts: learned templates need not be measurable in local probe rows alone, and a per-node bound by the full residual norm cannot be summed over all tree nodes as though there were only L of them. The actual comparator source includes the signed inherited return as well as the current residual commutator.

Several corrections are needed even before those counterexamples. The internal predesign is $16k \times 64k$, not the retained $8k \times 64k$ design. Weak directions must be indexed by the smallest singular values, and $d \leq s$ does not imply response rank d . A random inverse singular scale remains inside its joint expectation; conditioning on a probability-one full-rank event does not make it constant. Coefficients formed from $J(B)$ may themselves depend on exterior Ψ rows.

Finally, with $B_i = V_i^T J(B)$, $E_i = V_i^T J(E)$, $H = (B_1 + E_1)^\dagger$, and $H_B = B_1^\dagger$, the exact expansion is

$$Q(E) = E_2 - B_2 H_B E_1 + B_2 (H_B - H) E_1 - E_2 H E_1.$$

The last two terms cannot be omitted, and the adaptive V_i need not be comparator-only objects. These are additional algebraic and conditioning errors in the relayed proposal, distinct from its false screened-moment claim.

2.4 The old full-range learner: inverse growth and actual failure

Two different obstructions were established for the full-range commutator candidate with wide-query width $q = 64k$, response width $4k$, retained rank $r = 8k$, and physical leaves of size $16k$. This candidate retains the entire fitted response and chooses the remaining directions by inverse-optimal completion. It is not the maintained annihilated learner or the later synchronized union learner. Its all-depth exact-HSS rank and capture theorem remains valid; the noisy failure below does not contradict exact recovery.

2.4.1 An actual full inverse diverges while approximation is exact

For $k = 1$, let a, b be siblings inside a nonroot node p , and take the fixed one-way rank-one input $A = uv^T$, with u supported in a and v in b . Put $h = \Psi_a^T u$. On the b branch every nonzero response has left factor h , despite every outgoing physical block being zero. For a retained query space S_j , full-fit retention implies

$$P_{S_j} h = P_{\text{row}(G_j)} h.$$

Write $\delta_j = \|(I - P_{S_j})\hat{h}\|^2$. Conditional on h , the two phantom children use independent physical Gaussian rows and have axial symmetry about $\hat{h}$. Their mean residuals equal $d_\ell \hat{h}$, where $d_\ell = \mathbb{E}\delta_j$. Comparing the parent projection with the average of the two child projections gives

$$d_{\ell+1} \leq \frac{d_\ell + d_\ell^2}{2}, \quad d_0 = \frac{3}{4}, \quad d_\ell \leq 12 \cdot 2^{-\ell}.$$

The initial value concerns the full sixteen-dimensional Gaussian predesign: the adaptive eight-dimensional output preserves its fit.

The signal branch represents u exactly. If $D_a^T c_a = h$ and $D_b^T c_b = P_{S_b} h$, then $\|c_a\| = 1$ and

$$\sigma_{\min} \begin{pmatrix} D_a \\ D_b \end{pmatrix}^2 \leq \|h\|^2 \delta_b.$$

For a range-only tie and canonical-frame rule, scaling h does not change the selected spaces. Its Gaussian radius is therefore independent of δ_b , and

$$\mathbb{E}\|G_p^\dagger\|_{\text{op}}^2 \geq \frac{1}{(q-2)d_\ell} \geq \frac{2^\ell}{12(q-2)}. \quad (2.1)$$

All the designs in this particular finite-depth family are nonsingular: the child spaces projected onto $h^\perp$ have independent Haar orientations and dimensions whose sums are below $q-1$. This argument does not declare the selected Grams Wishart.

At p , however, both the outgoing block and response are zero. Inverse-optimal completion may discard the almost-canceling direction; its retained inverse trace satisfies $V_p \leq \min(V_a, V_b)$. Every true cut range is captured, and the fixed learned coefficient space contains A exactly.

Thus (2.1) refutes a uniform bound for the *full predesign inverse*, not approximation stability. The proof, exact scope of the range-only rule, and independent review are in `phantom_stability.md` and `phantom_stability_audit.md`.

An associated scalar shortcut also fails. For a normalized shared target, define $R = \beta/\|f\|^2$, where β is the query residual and f the minimum-norm physical fit. At a Gaussian leaf,

$$R \stackrel{d}{=} \frac{X_{q-m}X_{q-m+1}}{X_m}, \quad \mathbb{E}R = 168 \quad (m = 16, q = 64),$$

with independent chi-square variables. The actual parent law also depends on child metric matrices and their relative orientations. Reachable leaf states with an increasingly strong fitted direction and large optional directions give a limiting conditional mean exceeding 6/7 times the child ratio. Hence the proposed uniform conditional contraction by 6/7 is false; this does not rule out the weaker unconditional estimates subsequently used below. See `phantom_ratio_route.md` and `phantom_ratio_audit.md`.

2.4.2 A fixed-input lower bound for the entire retained projector

The later construction controls all seven optional directions, not only the mandatory fitted line. For small $\varepsilon > 0$, set

$$t = \left\lceil \frac{7}{2} \log(1/\varepsilon) \right\rceil, \quad \ell = t + 1, \quad N = 128 \cdot 2^t, \quad L = t + 3.$$

Two physical leaves of size sixteen form a source a_0 . Extend it through t active/blank merges to a ; let b be its sibling and $p = a \cup b$ a root child. With $u = e_1$, v_b a fixed unit vector uniform on b , and v_{ext} an exterior coordinate, define

$$B = \frac{u(v_b + v_{\text{ext}})^T}{\sqrt{2}}, \quad E_0[a_1, a_2] = E_0[a_2, a_1] = I_{16}, \quad A_\varepsilon = B + \varepsilon E_0.$$

All remaining entries of E_0 vanish. The matrix is fixed before sampling, B is globally rank one, and $\text{OPT}_1(A_\varepsilon) \leq 32\varepsilon^2$. For the full actual retained rank-eight space U_p , the reviewed result is

$$\mathbb{E} \text{dist}(u, U_p)^2 \geq c\varepsilon^{1.87}. \quad (2.2)$$

Since $A_\varepsilon[p, p^c] = uv_{\text{ext}}^T/\sqrt{2}$, every approximation confined to these learned nested row spaces incurs at least half this expected squared loss. Consequently its ratio to $L \text{OPT}_1(A_\varepsilon)$ diverges. Refitting or reducing rank inside the same spaces cannot repair this failure.

The proof has three substantive layers. First, a fixed compact source event of probability bounded below gives an initial query gap $\alpha_0 \asymp \varepsilon^2$ and a physical fit within $O(\varepsilon)$ of the represented signal. The active/blank physical martingale has $O(\varepsilon^2)$ squared error uniformly in its length. Its gap is α_0 times t independent Beta(24, 4) factors. Since their mean negative logarithm is less than 4/25, a high-probability state event satisfies

$$c_\alpha \varepsilon^2 e^{-(4/25)t} \leq \alpha \leq C_\alpha \varepsilon^2.$$

For the phantom sibling, the separately proved moment estimates give on a high-probability event

$$\beta \leq .439^\ell, \quad \phi_b \leq .699^\ell, \quad V_b \leq (9/8)^\ell, \quad R_b \leq (22/25)^\ell.$$

Here ϕ_b is the mandatory inverse cost, V_b the full retained inverse trace, and R_b the innovation-to-fit ratio. The fit-decay bound uses an exactly certified ninth-moment angular estimate, not a Monte Carlo estimate. The state event also bounds the total physical probe Frobenius norm by a constant times N .

Second, condition on the complete active state and phantom data invariant under query rotations fixing h . Eliminate the optional query coordinates and expose their planes and the remaining radii. On a further event of conditional probability at least $3/8$, a single relative angle c remains, with sphere-coordinate density $g_{49}(c) \propto (1 - c^2)^{23}$. Physical backsubstitution locates a zero c_* of the *actual physical overlap*; a zero of a projected query overlap alone would not suffice. On the interval

$$|c - c_*| \leq \kappa \frac{\sqrt{\alpha}}{\sqrt{T_\ell}}, \quad T_\ell = C_T(2 \cdot .699)^\ell,$$

the mandatory physical overlap is small and the parent mandatory inverse cost is at most a constant times ϕ_b .

Third, the Schur-energy estimate controls every inverse-optimal padding direction. Its vanishing terms include

$$\varepsilon^2[(2 \cdot .699)(9/8)]^\ell, \quad [2 \cdot .699^2]^\ell,$$

whose exponents are favorable. With κ fixed sufficiently small, the full retained space captures at most one quarter of the squared norm of u throughout the interval. Its conditional probability is at least a constant times $\sqrt{\alpha}/\sqrt{T_\ell}$, hence at least

$$c\varepsilon^p, \quad p = 1 + \frac{7}{4} \left(\frac{4}{25} + \log(1.398) \right) = 1.86632462667 \dots < 1.87.$$

All physical-fidelity and state events are imposed *before* the final shrinking angle interval. Subtracting an unrelated unconditional failure probability from that interval would not prove the claimed lower bound.

The complete proof and two independent reviews are recorded in `full_projector_bridge_obstruction.md` and `full_projector_bridge_audit.md`; its ingredients are `active_blank_martingale.md`, `phantom_negative_moment_route.md`, `phantom_negative_kernel_bound.md`, `physical_bridge_zero.md`, and `optional_cost_exclusion.md`. The theorem concerns exact full-range selection with the specified completion. An absolute singular-value cutoff can change that algorithm at small noise. The earlier $\varepsilon^{7/5}$ result concerned only its mandatory line; it must not be substituted for the full-projector proof.

The maintained annihilated learner survives this particular source mechanism. Its phantom sample vanishes and its blank frames are deterministic. For the stated leftmost two-leaf source family, with zero-outgoing siblings and vanishing perturbation commutator above the source, the reviewed bound is

$$\mathbb{E} \text{dist}_F(A, \mathcal{S}_U)^2 \leq \frac{191}{11} \|E\|_F^2 \leq \frac{764}{33} \text{OPT}_k(A).$$

These support conditions are essential; this is neither a general stability theorem nor evidence against (2.2). See `maintained_internal_source_bound.md` and `canonical_blank_path_audit.md`.

2.5 Separate views can lose a clean parent signal

Post-training union of independently propagated views does not ensure that a signal contained in the child union survives the parent union. The initial oblique example established this only for prescribed local states. The later construction reaches it through the actual maintained and raw two-view rules, including physical leaves, shared probes, column nullifiers, and canonical completion.

Take $N = 32$, $k = 1$, retained rank $r = 2$, and $q = 16$, with physical leaves of size four. Let a sixteen-coordinate root child I have children a, b of size eight. On the two leaves of a

choose orthonormal coordinate pairs $(u_1, p_1), (u_2, p_2)$ and put $u = (u_1 + u_2)/\sqrt{2}$. On b choose coordinates $b_1, \dots, b_4$; let f be exterior to I . The fixed input is

$$B = uf^T, \quad E = p_1 b_1^T + p_2 b_2^T, \quad A_\varepsilon = B + \varepsilon E.$$

The perturbation is entirely internal to I , so $J_I(E) = 0$. For $0 < \varepsilon < 1$, the singular values of the cut at a are $1, \varepsilon, \varepsilon$, and

$$\text{OPT}_1(A_\varepsilon) = 2\varepsilon^2.$$

Thus the comparator is best, rather than merely a convenient rank-one matrix.

The explicit full-rank probe witness in `two_view_union_reachability_route.md` produces child row planes

$$A_a = \text{span}\{(u + p_1)/\sqrt{2}, p_2\}, \quad R_a = \text{span}\{(u + p_2)/\sqrt{2}, p_1\},$$

and $A_b = \text{span}(b_1, b_2)$, $R_b = \text{span}(b_3, b_4)$. Consequently the child union contains u exactly. The child sample covariances on (u, p_1, p_2) can be taken as

$$S_{\text{ann}} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad S_{\text{raw}} = \begin{pmatrix} 6 & 0 & 2 \\ 0 & 8 & 0 \\ 2 & 0 & 6 \end{pmatrix}.$$

Their selected projectors have strict retained/discarded gaps. The single probe construction realizes both covariances and the required physical-leaf and column frames simultaneously; they are not prescribed as independent fictitious responses.

At I both actual row samples have structural rank one. Their lifted fits have the forms $A_1(u + p_1)/\sqrt{2} + B_1 b_1$ and $A_2(u + p_2)/\sqrt{2} + B_2 b_4$, with all displayed coefficients nonzero. In each canonical child-template coordinate order, the first a column is therefore independent of the mixed fit. It is used for padding before the second a column. The resulting spaces are

$$A_I = \text{span}(u + p_1, b_1), \quad R_I = \text{span}(u + p_2, b_4).$$

Their union projects u to $(2u + p_1 + p_2)/3$, and hence

$$\text{dist}(u, A_I + R_I)^2 = \frac{1}{3}. \tag{2.3}$$

The current residual commutator is zero, the parent outgoing comparator is rank one, and neither view truncates a nonzero parent sample tail. The loss is inherited through the separately propagated templates.

This is an actual-recursion reachability statement. For each fixed $\varepsilon > 0$, full design ranks, spectral gaps, nonzero mixing coefficients, and the relevant first canonical pivot persist on a positive-probability Gaussian neighborhood. Only the needed physical subspaces and first a pivot must be continuous. Some unused canonical columns can change gauge discontinuously, and the proof does not assume otherwise. The parent rank-one sample is structural under probe perturbations, rather than a nongeneric zero singular value imposed by the witness.

The witness uses narrow-probe coordinates of size $1/\varepsilon$. Its neighborhood probability may depend on ε ; (2.3) does not by itself yield an asymptotic expected-error lower bound. Independent deterministic evaluations of the actual dense learners at $\varepsilon = 1, .1, .01$ reproduce $1/3$ and full rank-four predesigns at all fifteen nodes. The algebra and continuity argument, not those evaluations, prove reachability. See `two_view_union_reachability_audit.md` and, for the earlier unreached local example, `two_view_union_oblique_route.md`.

The synchronized repair and its limits. Using the same child-union templates for both parent views removes this exact mechanism. The specified synchronized learner retains at most $4k$ directions with width $32k$, retains the full raw fit on its low-rank branch, and otherwise combines the lifted raw leading $2k$ directions with the annihilated fit. Its completion first fills query visible directions and then physical kernel directions if necessary. For exact HSS- k and the stated rank-slack inputs, rank and true-cut containment have proofs. A clean parent incurs no new loss when its common child template contains the comparator and its predesign is injective on the relevant coefficient space; on singular branches an invisible comparator component must be retained explicitly in the identity. Full rank of a retained design is weaker than full rank of the next stacked predesign. None of these statements provides the joint inverse-weighted budget for arbitrary noisy signed sources. See `synchronized_union_exact_route.md`, `synchronized_union_route.md`, and their audits.

2.6 Conditioning, passive memory, and changing targets

2.6.1 What the renewed latent law permits one to condition on

For the independent-source fitted-target tree, the exact innovation mixture is a theorem for a restricted recursively generated summary. It may retain physical bases and fits, allowed local and retained Grams, and deterministic functions of these data. It may not retain forgotten query-coordinate designs, old residual vectors or radii, or arbitrary response cross-Grams. Selection must be invariant under right rotations of query coordinates and must contain the entire fit.

For example, after temporarily exposing two current child designs and their scalar latent variances τ_a, τ_b , the parent residual before compression has covariance

$$(\tau_a + \tau_b)N_G,$$

where N_G projects onto the kernel of the full $2r$ -row predesign. The fitted increment and this residual are conditionally independent Gaussians. The residual is not isotropic in the larger kernel of the retained r -row design under that same conditioning. Only after forgetting the child query frames and using the retained-space stabilizer can one replace its direction by a uniform direction in that larger kernel. The beta-chi-square identity then supplies

$$\tau_{\text{new}} = (\tau_a + \tau_b) Z, \quad Z \sim \text{Beta}\left(\frac{q - 2r}{2}, \frac{r}{2}\right)$$

in the reduced realization. This is not a license to restore all the discarded records afterwards. Two explicit pitfalls illustrate the restriction. A selector that inspects a fixed query coordinate can orient an optional direction toward that coordinate, contradicting the alleged isotropic residual variance. Even an invariant record can be excessive: retaining the old residual radius, the new fit, and its Gram determines the new radius by subtracting the fitted energy. A fresh nondegenerate beta radius is then impossible. Right-rotation invariance alone is therefore insufficient. The corrected theorem and its examples are in `fitted_target_exact_audit.md`; the matrix version uses the same restricted-summary obligation in `matrix_latent_budget.md`.

The source posterior does not renew as a Wishart posterior at every merge. With block pre-Gram $K = \begin{bmatrix} A & C \\ C^T & B \end{bmatrix}$ and Schur complements $S_a = A - CB^{-1}C^T$, $S_b = B - C^T A^{-1}C$, the fitted increment covariance is

$$\Omega = K^{-1} \text{diag}(\tau_b S_a, \tau_a S_b) K^{-1}.$$

Its likelihood involves

$$\frac{(Kz)_a^T S_a^{-1} (Kz)_a}{\tau_b} + \frac{(Kz)_b^T S_b^{-1} (Kz)_b}{\tau_a}.$$

Already on a scalar cross-block slice this is proportional to $(\tau_b^{-1} + c^2\tau_a^{-1})/(1 - c^2)$, not a linear precision term in K . Source conjugacy thus does not establish learned-Gram conjugacy. Likewise an inverse-weighted residual cost contains the actual product $\mathbb{E}[(\tau_a + \tau_b)V_p]$; replacing it by a product of expectations drops its covariance term. See `binary_posterior_closure.md`.

These obstacles did not prevent every later local theorem. The finite moment ladder bypasses conjugacy, retaining the dependent products and conditional Haar orientations. With its stated independent-source assumptions, $r \geq 4096$, $q = 8r$, and target width at most $r/2$, it establishes uniform directional error moments and inverse means with large absolute constants. It is not a theorem for $r = 2$, nor an automatic interface to actual exterior HSS commutators. See `fitted_energy_bootstrap_route.md` and `fitted_energy_bootstrap_audit.md`.

2.6.2 Projection loss, fitted energy, and passive records differ

A physical projection error can be bounded by the target norm on a bad event. A fitted coefficient can be arbitrarily large on that same event and affect ancestor selections. Thus paying only the local projection loss does not dispose of the future fitted-energy cost. For the fitted-target martingale $S_t = F_t - C_0$, a valid global stopping argument with exit threshold $M\|C_0\|$ gives

$$\mathbb{E}\|E_H\|^2 \leq \beta + (1 + M^{-2}) \mathbb{E}\|S_{H \wedge \tau}\|^2,$$

where the stopped quadratic variation includes the exit increment. Discarding that overshoot changes the process. A global fit bound does not normalize every small local target; conditioning on survival can also couple siblings. The reduction in `fitted_target_stopping.md` was valid but did not close its remaining bound. Later large-rank source theorems should not be read back into that scalar stopping argument.

Merely carrying an old target or error vector as a passive record does not make it an actively retained fit. The next compression may discard it, after which its update has an additional projection loss and is not the centered full-fit recurrence. Even retaining each current joint and separate child fit need not preserve older witnesses. `augmented_memory_route.md` gives a full-rank two-merge local construction in which current fits and inverse-optimal padding eventually exclude a nearly represented signal. That construction was checked as an admissible local state, not reached from the actual Gaussian leaf process; it is not an expected-error counterexample. The joint-plus-split learner nevertheless has a valid exact-input endpoint at $r = 16k$, $q = 64k$, response width $4k$, and physical leaves of size $32k$. Its three mandatory fits have combined dimension at most $12k$, and the exact-input boundary-anchor induction proves rank and cut containment. A clean rank-one bridge retains a good separate fit from either child. That one-bridge protection does not automatically protect the same vector at the next ancestor. See `split_fit_learner_audit.md`.

Nor does physical closeness to a fixed envelope suffice. If $Q = DC + E$, $D^T E = 0$, and $\|E\|_{\text{op}} \leq \delta < 1$, then

$$\sigma_{\min}(Q^T X) \geq \sqrt{1 - \delta^2} \sigma_{\min}(D^T X) - \delta \|(I - DD^T)X\|_{\text{op}}.$$

This gives good-event inverse moments under a *relative probe-space* perturbation condition. Without it, for $n > q$ an adaptive unit vector arbitrarily close to a fixed e can approach $\ker X^T$ on a positive-probability event while remaining nonsingular. Its inverse cost grows like an arbitrary inverse squared parameter. An accompanying fixed-target example has zero query innovation but arbitrarily large fitted coefficient. It can even converge eventually to the exact core for every realization while its expectations diverge. These are right-rotation-invariant adaptive constructions, not reached nested-tree examples. They refute a bridge based only on geometric closeness and explain the need for uniform integrability. See `near_envelope_route.md`.

A related historical capture obstruction is also limited in scope. The marginal facts $D = T - B$, $T/\sigma^2 \sim \chi_{15}^2$, small L^2 residual B , and $D \geq_{\text{st}} \chi_2^2/2$ alone permit abstract couplings with an infinite first inverse moment. Such couplings need not satisfy the actual fresh latent-radius law. Once that law is kept, the later Laplace argument gives $\mathbb{E}D^{-p} \leq \sigma^{-2p} 2^{-p} \Gamma(3/2 - p)/\Gamma(3/2)$ for $p < 3/2$, including $\mathbb{E}D^{-1} \leq \sigma^{-2}$. Thus the marginal construction is not an actual-tree counterexample. Compare `actual_full_response_capture.md` with `latent_capture_laplace.md` and its independent audit.

2.6.3 Target drops require a retained deterministic guard

Deterministic compatible transfers $U_I|_i = U_i A_i$ preserve the additive physical target and transform omission covariances by congruence, $C_I = \sum_i A_i^T C_i A_i$. Invertible changes of output coordinates preserve the relevant fit ranges. A strict drop is different: a fixed physical prefix that would complete the *current* smaller target may already have been discarded while an earlier larger target was being fitted.

The explicit two-target source example has a fixed prefix of length $r-2$ and a random fitted two-plane. A prescribed next coordinate almost surely lies outside the retained space. Dropping one output at the parent does not recreate that coordinate. Treating its learned replacement as a fixed Gaussian direction is invalid. Without a history-compatible prefix condition, the inverse decomposition has an extra nonnegative padding cost

$$\mathcal{P} = \|B_{\text{phys}}(I - P_{\text{range}} Z)\|_F^2,$$

which persists when there are more unexcited available directions than active target columns. The moment recurrence must then charge inverse-weighted products involving $\mathcal{P}$. See `nested_target_bootstrap_route.md`.

The fixed-guard modification repairs this particular problem. With $s_{\text{max}} \leq r/2$, put $c = r - s_{\text{max}}$. Each source has a deterministic pool of $2c$ directions and retains its first c as a guard. At a parent, the block sum of its two child guards is an available deterministic pool of $2c \geq r$ directions; select the full fit and complete by scanning this pool. Since the fit rank is at most s_{max} , the first c pool directions remain retained. This is an implementable induction, not an oracle reopening a discarded physical coordinate. The enlarged-core argument then restores the original local moment constants under arbitrary compatible deterministic drops. See `fixed_guard_completion_route.md` and `fixed_guard_completion_audit.md`. It repairs a local source tree; independent innovations for arbitrary HSS assembly remain a separate requirement.

2.7 Exterior information and pilot independence

2.7.1 Comparator anchors do not describe actual residual records

At a fixed cut I , all exterior *comparator* anchors restrict inside I to its rank- k outgoing space. Conditioning on those anchors removes at most k local Gaussian directions. Descendant anchors are not included in this statement. An orthogonal comparator wavelet decomposition also does not make children conditionally independent: if

$$\begin{pmatrix} X_a \\ X_b \end{pmatrix} \Big| X_I = \begin{pmatrix} R_a \\ R_b \end{pmatrix} X_I + QG,$$

then $\text{Cov}(X_a, X_b \mid X_I) = -R_a R_b^T$ in normalized coordinates. Fixing a parent sum creates a contrast coupling.

Actual exterior sketches can expose arbitrarily more directions, even with a best comparator. At a root cut K, J , take fixed rank- k frames U, V and orthonormal leaf-supported directions $P = (p_1, \dots, p_M), Q = (q_1, \dots, q_M)$ orthogonal to the respective comparator frames. Set

$$B = \sigma(UV^T + VU^T), \quad E = \varepsilon PQ^T, \quad 0 < \varepsilon < \sigma.$$

On the specified physical-leaf/coarse tree, B is HSS- k , and the root-cut singular values give $\text{OPT}_k(B + E) = M\varepsilon^2$ exactly. Fix the narrow probe, exterior wide rows, and comparator anchor. Each actual exterior leaf commutator then reveals $\varepsilon h_j x_j$, where $h_j = \Psi_K^T p_j$ and the observed row $x_j = q_j^T \Omega_{J_j}$ is nonzero almost surely. It therefore reveals every h_j . The remaining local physical Gaussian covariance is $I - UU^T - PP^T$, losing M residual directions, not just $O(k)$. This is a conditioning obstruction, not an approximation lower bound. See `correlated_source_interface_route.md` and its audit.

The same distinction applies to internal injections. Deterministic additions in the currently retained guard pool are reproduced exactly and preserve the local theorem. A genuinely omitted addition is fresh only if its physical query directions were reserved from the entire previous computation, including target responses and conditioning records. Orthogonality to the final selected template alone does not give that independence. Ordinary binary parent rows already belong to the children; comparator contrasts of dimension at most $2k$ do not automatically supply unused Gaussian coordinates. Even a valid reserved-source model must charge injections by their ages, since recent injections have not undergone many residual contractions. See `guarded_internal_injection_route.md`.

2.7.2 Independent pilots leave exterior dependence and width costs

Freeze an independently learned rank- r column pilot V_I , then use fresh probes and its annihilator N . With $R = A[I^c, I](I - V_I V_I^T)$ and annihilator width s ,

$$Z = \Psi_{I^c}^T R \Omega_I N, \quad \mathbb{E}(\|Z\|_F^2 \mid V_I) = qs \|R\|_F^2.$$

For a fixed m -dimensional local template the corresponding lifted mean is $ms \|R\|_F^2 / (q - m - 1)$. These are useful exact one-step costs. But a fixed incoming cut of rank $r + 1$ defeats every rank- r pilot. Conditioning on that pilot and the local design, a nonzero surviving incoming response produces a Gaussian fit with full physical covariance. A retained space containing it cannot be measurable in local fresh rows alone. Exposing a common exterior anchor subsequently correlates siblings. Independence of the pilot from the fresh probes does not prove local independence of the *new learned* hierarchy. See `crossfit_hss_interface_route.md` and its audit.

An oracle collection of independent full-row exterior pilot ranges is also not automatically a bounded-width nested envelope. Along a path, fixed shell perturbations can add p mutually orthogonal directions at each of h levels. Differences of ancestor sketches expose each new block through an invertible $p \times p$ Gaussian factor, forcing dimension at least hp in the nested closure at the terminal node. Conversely, intersecting a clean parent signal envelope with a noisy child range can yield the zero space. Projecting that signal into the child range restores feasibility but reintroduces adaptive dependence. These facts in `pilot_route.md` do not exclude all pilots; they exclude those proposed exact-envelope constructions.

2.7.3 Clean full-row linear extraction costs depth

Let $W \in \mathbb{R}^{N \times M}$ collect every forward query. An input-independent linear extraction of a clean exterior sketch at I has the form $A[I, :] W K_I$ with $W[I, :] K_I = 0$. On a path $I_0 \supset \dots \supset I_L$,

choose a fixed rank- k test block between I_d and k coordinates of its sibling. Exact capture of that test forces

$$\text{rank } W[I_{d-1}, :] - \text{rank } W[I_d, :] \geq k.$$

Summation gives $M \geq kL$. Only finitely many fixed test inputs are needed for the simultaneous probability-one assertion. Adding transpose queries of width M_H does not evade it on nodes larger than M_H : an invisible perturbation $(I - P_H)T(I - P_W)$ shows that exact extraction either has its right factor in range W , or requires the entire coordinate space of I in range H .

Finite sign averaging at one partition has the additional problem that its variance is comparator-sized, even when the approximation residual is zero; finite Rademacher masks can erase a fixed rank-one interaction with positive probability. The lower bound concerns clean *full-row*, input-independent linear extraction. The two-sided compressed commutator already available with $O(k)$ probes lies outside that model, as do nonlinear and adaptive decoders. See `tree_mask_route.md`.

2.8 Frozen compression and shared external noise

For independent pilot frames P, Q and fresh Gaussian probes, frozen compression has the exact conditional law

$$Y_c = A_c \Omega_c + N_y, \quad X_c = A_c^T \Psi_c + N_x, \quad A_c = P^T A Q,$$

where the four arrays $\Psi_c, \Omega_c, N_y, N_x$ are initially independent. The noise columns have fixed, possibly dense physical covariances C_y, C_x , and

$$b = \text{tr } C_y + \text{tr } C_x \leq \|A - PP^T A QQ^T\|_F^2.$$

If the pilot contains a comparator, $b \leq \|E\|_F^2$. At each complete level the expected commutator-noise energy is exactly $q_\psi q_\omega b$. These are orthogonal energy identities, not independence between nodes; reusing the probes in learned subsequent compressions does not renew the four-array law. See `frozen_compression_noise_route.md` and its audit.

Factor $C_y = L_y L_y^T$, $C_x = L_x L_x^T$. Every actual core computation can be realized simultaneously as an exact commutator computation for

$$A_{\text{ext}} = \begin{pmatrix} A_c & L_y \\ L_x^T & 0 \end{pmatrix}, \quad \Psi_{\text{ext}} = \begin{pmatrix} \Psi_c \\ G_x \end{pmatrix}, \quad \Omega_{\text{ext}} = \begin{pmatrix} \Omega_c \\ G_y \end{pmatrix},$$

with independent standard Gaussian exterior arrays. The external perturbation has squared Frobenius norm b . This pathwise realization preserves the actual core recursion; its root is a proper node of the augmented problem, not a new zero-commutator global root.

Rank-one incoming noise already defeats the independent-source product law. Set $A_c = N_y = 0$ and $N_x = v h^T$. The physical-leaf row frames are deterministic, so two first internal sources have independent Gaussian M -row designs G_a, G_b but the same h . Their residuals $e_i = (I - P_{G_i})h$ obey

$$\begin{aligned} B_i &= \|e_i\|^2 = \|h\|^2 Z_i, \\ Z_a, Z_b &\text{ independent Beta}((q - M)/2, M/2), \\ \text{Cov}(B_a, B_b) &= \frac{2(q - M)^2}{q} > 0. \end{aligned}$$

Each marginal is the expected chi-square law. At $q = 8r$, $M = 2r$ the covariance is $9r$. Exposing the complete source designs and fits gives $h = \mu + \eta$, where, conditionally on the designs,

$\eta \sim N(0, I - P_{[G_a; G_b]})$ is independent of the fits. The next retained design R lies in the old stacked row space, so

$$\|(I - P_R)h\|^2 = \|(I - P_R)\mu\|^2 + \chi_{q-2M}^2$$

under this complete record. Ignoring the coherent mean is incorrect. This stronger conditioning is not a contradiction to the restricted latent theorem; the independent-source hypothesis already fails by the positive unconditional covariance. Nor does the example prove physical instability, since its core matrix is zero. See `independent_noise_tree_route.md` and its audit.

There is a positive rank-slack exception. If augmented outgoing and incoming cut ranks fit within the maintained retained rank, its own hybrid column hierarchy captures the relevant external incoming factors, and its nullifiers remove them. In particular a rank-one incoming external perturbation of an exact HSS- k core can leave the maintained row trajectory identical to the noiseless trajectory. The hybrid's own column frames are essential; substituting raw companion frames does not prove the assertion. Low covariance trace alone does not imply these rank inequalities. See `two_view_external_noise_route.md` and its audit.

The noncentral projector theorem supplies another one-step estimate, without pretending inherited noise is fresh. For fixed M , standard Gaussian $Z \in \mathbb{R}^{t \times s}$, $s < t$, and $U = \text{range } M$,

$$\mathbb{E}P_{M+Z} \preceq P_U + \frac{s}{t}P_{U^\perp}.$$

The coefficient s/t is sharp; singular covariance is handled by quotienting out deterministic mean-only columns before whitening. There is no general opposite inequality $\mathbb{E}P_{M+Z}|_U \succeq (s/t)I_U$: for $t = 3, s = 2, M = [ae_1, \delta e_2]$, first large a then small positive δ gives limiting leverage $1/2$ on e_2 , below $2/3$. In a guarded injection this upper bound interpolates an inherited-fit metric cost with a full pretemplate cost. It does not bound the inherited cost or close arbitrary mixed histories. See `noncentral_projector_route.md` and its audit.

2.9 Consistency correction and global correction barriers

2.9.1 Consistent observations can require large matrix noise

Noisy compressed products generally violate $\Psi^T Y = X^T \Omega$. Orthogonal least-squares repair is explicit. With $S = \Psi \Psi^T$, $T = \Omega \Omega^T$, its minimum-norm matrix representative solves

$$SM_* + M_*T = \Psi X^T + Y \Omega^T.$$

If the noise-free core is A , then

$$M_* = A - (I - P_\Psi)A(I - P_\Omega) + Z, \quad SZ + ZT = \Psi N_x^T + N_y \Omega^T.$$

The repaired observations are products of $A + Z$. The accurate witness $A + Z$ need not be computable from underdetermined observations; the computable M_* loses the invisible block. These two matrix objects must not be identified.

In eigenbases of S, T , conditional noise energy equals

$$\mathbb{E}(\|Z\|_F^2 \mid \Psi, \Omega) = \sum_{\lambda_i + \mu_j > 0} \frac{\lambda_i c_{x,j} + \mu_j c_{y,i}}{(\lambda_i + \mu_j)^2}.$$

Away from square dimensions, inverse-Wishart moments give a bound by b when each larger dimension is at least twice the smaller plus one. Near-square behavior is different. With $n_r = 2q$, $n_c = q + 2$, both query widths q , and $C_x = 0$, the mean correction energy is at least $qb/2$. A physical nullspace on the opposite side and dimension gap at most one gives an infinite

mean. In the doubly square case the mean is also infinite: the smallest two eigenvalues are bounded above by independent χ_1^2 variables, whose sum has an infinite inverse mean. This defeats an unqualified stable orthogonal consistency repair, not every regularized estimator. Even where finite, Z depends on the same designs and cannot be fixed while those designs are subsequently treated as fresh. See `compression_consistency_repair_route.md` and its audit.

2.9.2 A few global low-rank stages need a global-tail premise

For a known pilot M_0 , fresh range queries on $A - M_0$ and adapted transpose queries give an implementable correction

$$M_1 = M_0 + QQ^T(A - M_0), \quad \mathbb{E}(\|A - M_1\|_F^2 \mid M_0) \leq \left(1 + \frac{k}{p-1}\right) \tau_k(A - M_0),$$

where Q comes from a Gaussian sample of width $k + p$, $p \geq 2$, and τ_k is the squared global rank- k tail. For $p = k + 1$ the total new query cost is at most $4k + 2$. The missing premise is control of that *global* tail by $L \text{OPT}_k$.

HSS rank does not supply it: m disjoint unit rank-one sibling blocks are HSS rank one but have global tail $m - k$. A fixed number of rank- $O(k)$ corrections from zero cannot recover them when m is large. Similarly, with frozen P, Q ,

$$\|A - PCQ^T\|_F^2 = \|A - PP^T AQQ^T\|_F^2 + \|P^T A Q - C\|_F^2;$$

exactly solving the compressed core cannot improve the fixed geometric floor. Requerying full products at each level costs the rank of the concatenated lifted query spaces, which can be proportional to the number of levels. Preserving free leaf-diagonal blocks is also necessary: an identity matrix is HSS rank zero but has large global rank.

Even orthogonal projection of an HSS- k comparator onto an arbitrary fixed nested coefficient space need not retain HSS rank k : different levelwise projections can make formerly common row factors independent on one cut. Exact coefficient projection is not automatically available from small fixed transcripts either. An invisible-matrix test shows that exact projection onto a space containing every diagonal matrix requires $\dim \text{range } \Psi + \dim \text{range } \Omega \geq N$. These statements do not invalidate the separately proved approximate independent refit, nor an offline best-HSS rank reduction once a good matrix approximation has actually been obtained. They distinguish query complexity from a fast or directly computable optimization rule. See `bounde_d_stage_compression_route.md` and its audit.

Unreviewed draft extension at the research stop. The final `replicated_global_tail_obstruction.md` was saved as a proved draft but had not received an independent audit. It should not be assigned the review status of the preceding results. Its pointwise compression argument claims that, for disjoint tested cuts with losses e_j and amplitudes a_j , every pilot obeying their learned row constraints satisfies

$$\tau_d(A - M_0) \geq \sum_j |a_j|^2 e_j - \sum_{\text{largest } d} |a_j|^2 e_j.$$

It proposes applying this to block-diagonal copies of the reviewed old full-range counterexample, obtaining a failure of any fixed number of global correction directions to restore an $O(L \text{OPT})$ tail premise for that learner. It does not target the synchronized learner or arbitrary nonlinear structured corrections. No new claim of independent verification is made here.

2.10 Numerical evidence and the final status map

The saved experiments test different objects and should not be pooled into a single claim. The old full-range proof-chain batch contained 1064 recursive cases up to $N = 65536$, with 32–160 independent draws per noisy cell. It identified the bridge mechanism and checked identities, but the later analytic rare-angle argument proves the $\varepsilon^{1.87}$ lower bound. A conditional rotation diagnostic on one child state is likewise not an unconditional expectation bound. The guarded full-range batch contained 800 cases from 160 independent probe pairs, reusing pairs across noise amplitudes. At $N = 131072$ and $\varepsilon = .1$, mean signal loss rose from .08175 to $.62016 \pm .05154$ across the bridge. This finite amplification neither refutes an $O(L)$ estimate nor tests the large-rank independent-source theorem’s hypotheses. See `proof_chain_numerical_summary.md`, `bridge_conditional_rotation.md`, and `guard_fullrange_bridge_check.md`.

The synchronized learner’s separate batch used 624 cases from eighty independent probe pairs. Its largest exact squared coefficient-space error was 3.162×10^{-23} ; its largest noisy-cell mean error divided by the *specified comparator residual* was $1.00792492 \pm .00335957$. The high-rank internal union branch was exercised 10752 times. The experiment used an oracle projection onto the fixed learned coefficient space, not the final random refit or a rank- k output, and did not compute best-HSS OPT. Reused seeds across families and amplitudes are paired cases, not extra independent draws. The follow-up signed diagnostic reused three probe pairs and checked 400 reached oriented nodes. Its maximum vector-identity discrepancy was 2.721×10^{-15} , with both substantial cancellation and amplification between the source and inherited-return terms. No expectation factorization follows from those checks. See `synchronized_union_numerical_check.md`, `synchronized_union_numerical_audit.md`, and `synchronized_signed_identity_check.md`.

The noncentral projector experiment is narrower still: fixed means, covariances, and positive semidefinite metrics with fresh Gaussian arrays. Its 1150000 evaluations include 20000 repeated noiseless deterministic evaluations, so only 1130000 are nonempty Gaussian draws. Scalar and central equality checks, singular quotient identities, and the anisotropic negative control agree with the proved formulas within their recorded numerical and sampling errors. It does not test freshness of inherited HSS noise or closure of the mixed-history cost. See `noncentral_projector_numerical_check.md` and its audit.

The endpoint can be stated without conflating these scopes:

1. **Defeated selectors and acquisition models.** Minimum-norm and the stated nuclear/tree penalties fail even with exact leaf blocks. The one-sided Gaussian pilot and clean full-row linear extraction have their stated lower bounds. None is an impossibility theorem for all nonlinear two-sided decoders.
2. **A defeated actual learner.** The old inverse-optimal full-range candidate fails the desired expected physical bound. Its earlier mandatory-line padding gap was closed by the full-projector proof. The maintained learner’s positive theorem on that source family remains compatible with this result.
3. **A reached local defect repaired by synchronization.** Separate-view propagation can incur a clean-parent loss despite zero child-union loss. A common-template synchronized merge removes that specific defect under its visibility hypothesis. Its exact and rank-slack endpoints do not prove noisy physical stability.
4. **Local source issues with precise repairs.** Restricted conditioning restores the latent law; fixed guards repair historical target-drop padding; the finite moment ladder closes the sufficiently large-rank independent-source model. These repairs do not manufacture independent external commutator sources.
5. **Remaining coupled quantities.** Shared external noise, residual-dependent anchors, and the adaptive left lift remain coupled to inherited physical errors. The proved unlifted signed-

source and tail budgets must still be converted into a global expected physical budget without multiplying by an independently averaged inverse or double-counting cumulative losses.

6. **Reconstruction and the unresolved conjecture.** Independent fixed-space refitting and query-only rank reduction remain available under their proved premises. No documented result supplies the required all-input synchronized basis-quality premise. Numerical checks do not supply it, and the final replicated-tail note remains an explicitly unreviewed draft extension. The original general $O(k)$ -query, $O(L \text{OPT}_k)$ conjecture is unresolved.

Source provenance

The following local records supply the result statements, their assumptions, and the internal audits cited in this manuscript. They are provenance for the present research record, not public bibliographic publications. The assembled LaTeX file is self-contained. Filenames below identify the original supporting records in the project directory.

- active_blank_martingale.md
- actual_full_response_capture.md
- augmented_memory_route.md
- binary_posterior_closure.md
- bounded_stage_compression_route.md
- bridge_conditional_rotation.md
- canonical_blank_path_audit.md
- certified_selector_audit.md
- certified_selector_route.md
- claudescreened_residual_audit.md
- compression_consistency_repair_route.md
- correlated_source_interface_route.md
- crossfit_hss_interface_route.md
- fitted_energy_bootstrap_audit.md
- fitted_energy_bootstrap_route.md
- fitted_target_exact_audit.md
- fitted_target_stopping.md
- fixed_guard_completion_audit.md
- fixed_guard_completion_route.md
- frozen_compression_noise_route.md
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- full_projector_bridge_obstruction.md
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- guarded_internal_injection_route.md
- hss_audit_report.md
- hss_complete_proof_chain.tex
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